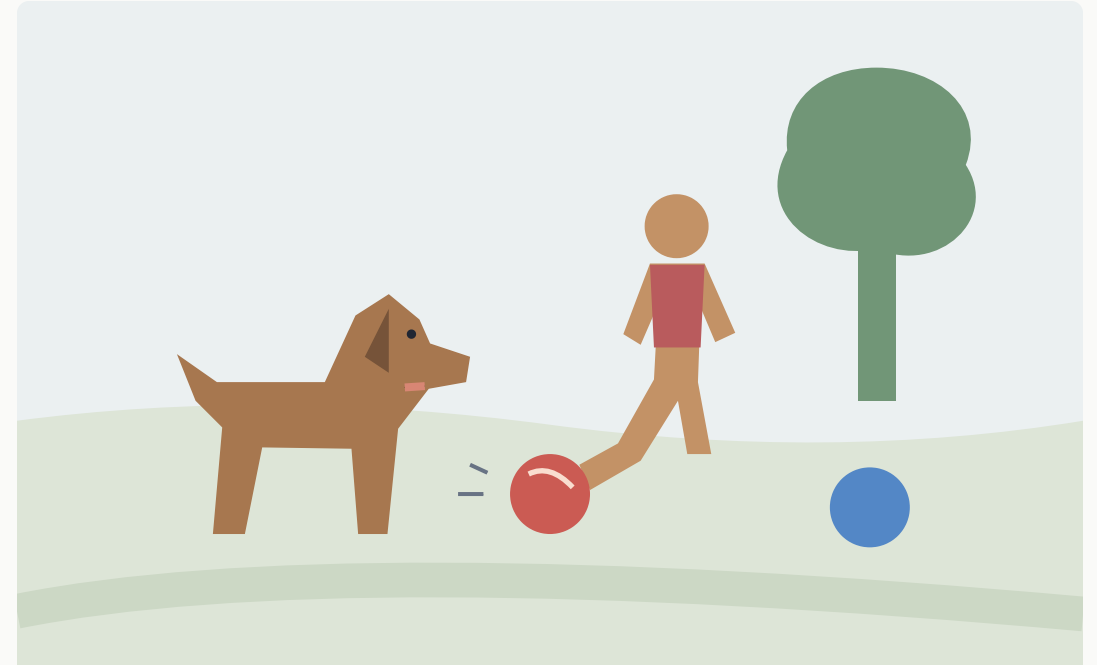


LANGUAGE-GROUNDED VISION

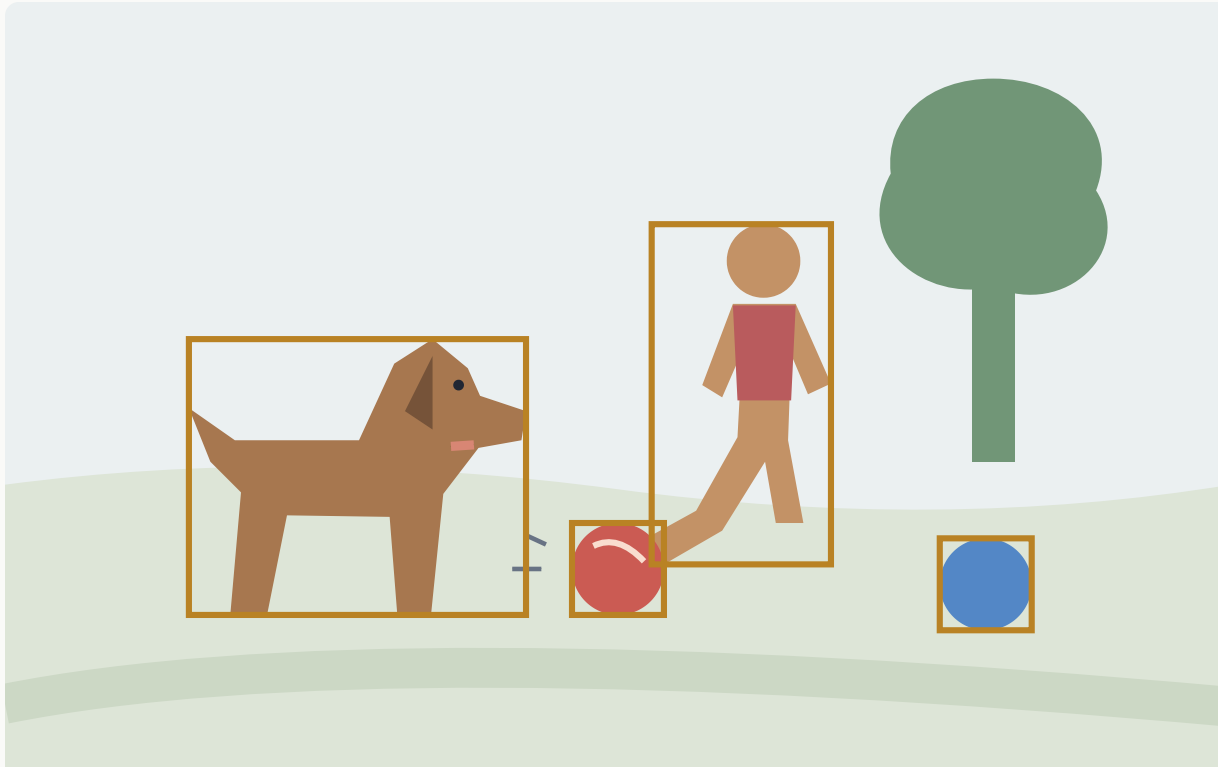
How Words Become Boxes, Regions, and Masks

HOW DOES LANGUAGE CHANGE
WHICH PART OF THE IMAGE WE
PREDICT?

ES 667 • Nipun Batra • 55-minute route



A detector already gives us locations



"dog"

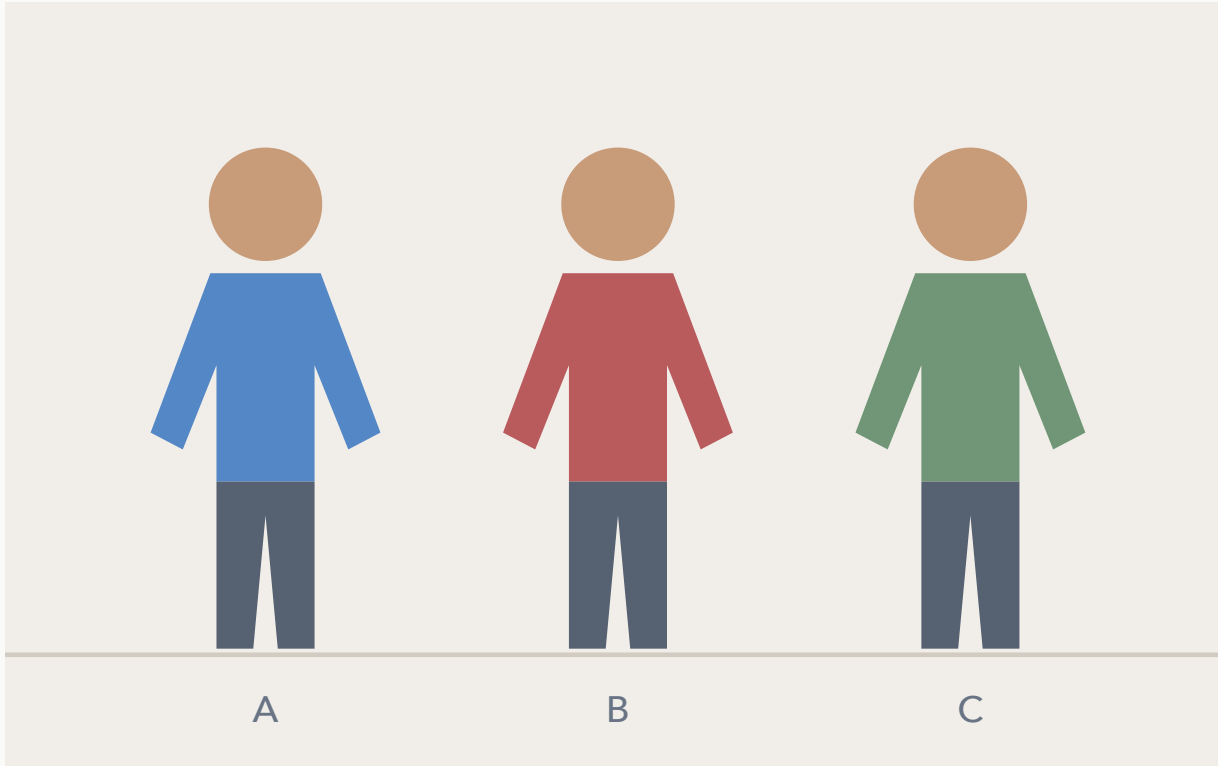
$$I \mapsto \{(\hat{\mathbf{b}}_i, c_i, s_i)\}_{i=1}^N$$

Box · class label · score

Familiar fixed-class detector

Where is already part of detection. Which target is still constrained by the label set.

"Person" does not identify one person



"person"

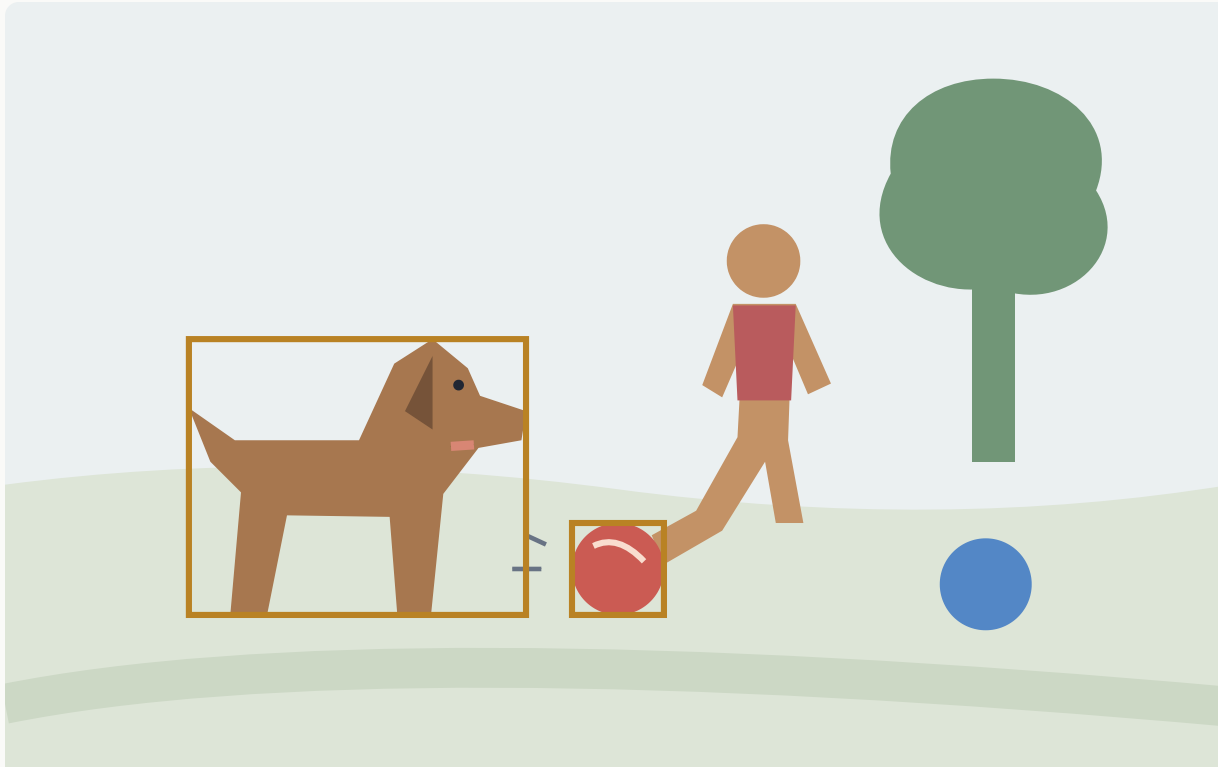
A, B and C all match.

"person wearing red"

B · the attribute selects an instance

| A class name is sometimes not enough.

Change the phrase; change the target



"red ball"

Attribute selects the red ball.

"ball beside the dog"

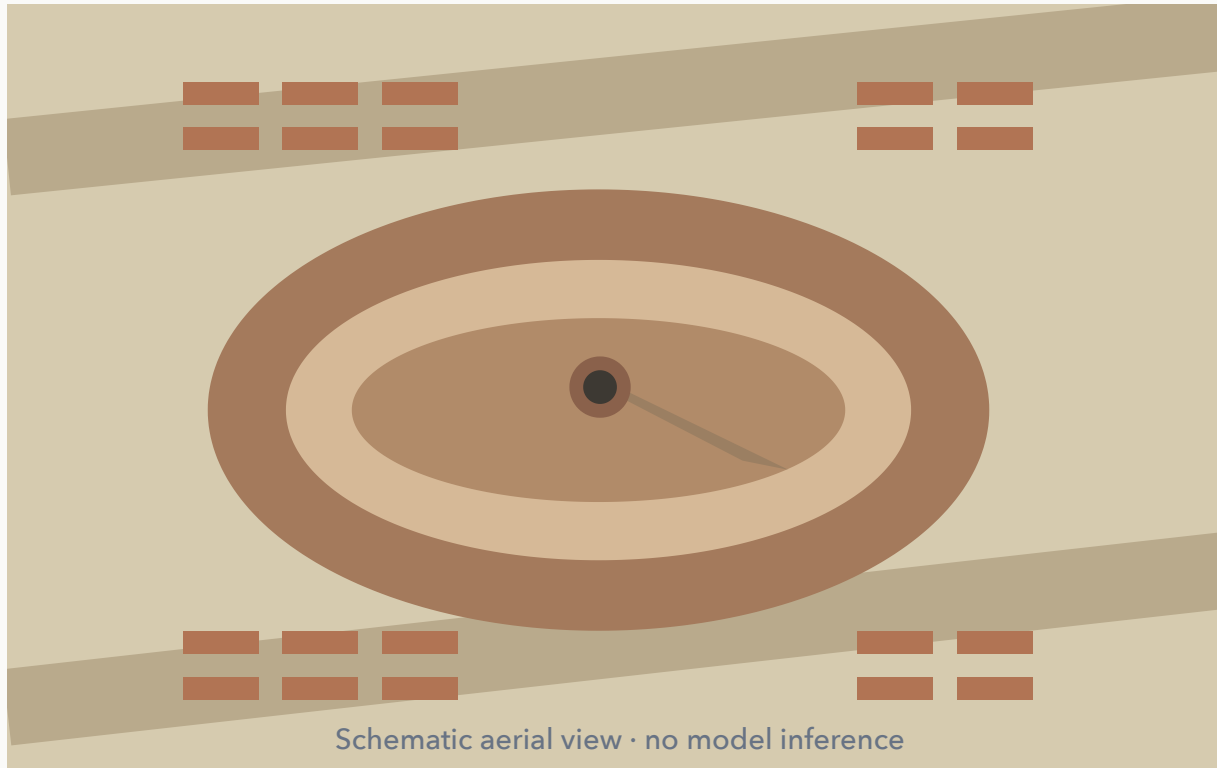
Relation selects that same ball.

"dog beside the red ball"

Now the requested object is the dog.

| The head noun names the target; the relation constrains it.

The words can ask for unfamiliar evidence



"find the kiln"

"find its chimney"

Same domain. Different spatial target.

Schematic aerial view.

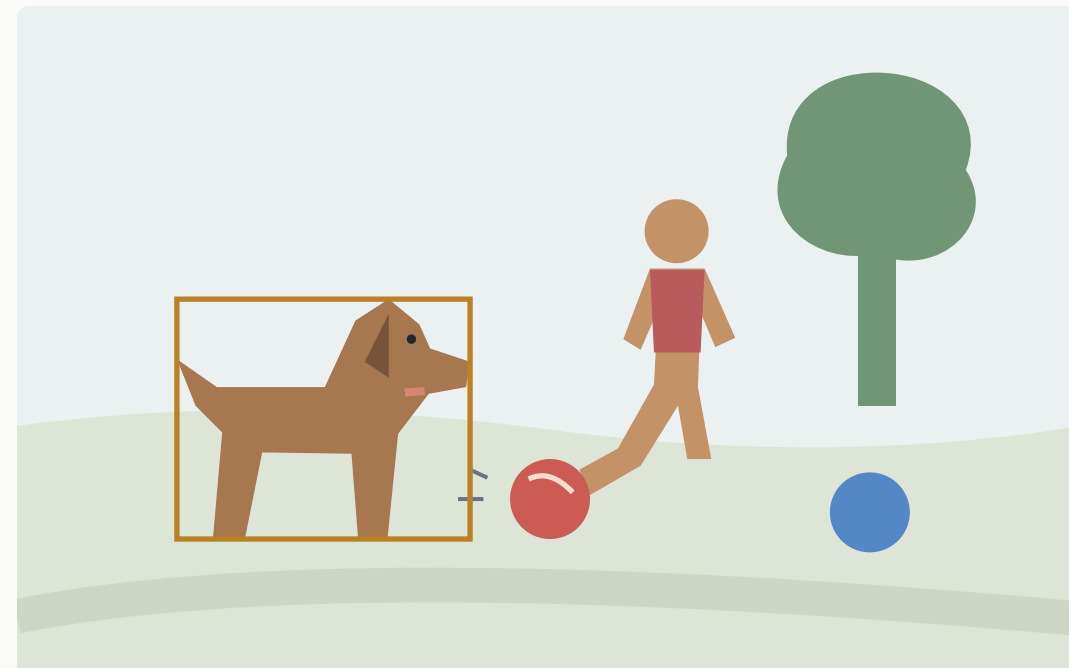
Not a satellite observation.

An open vocabulary does not remove domain shift or missing visual detail.

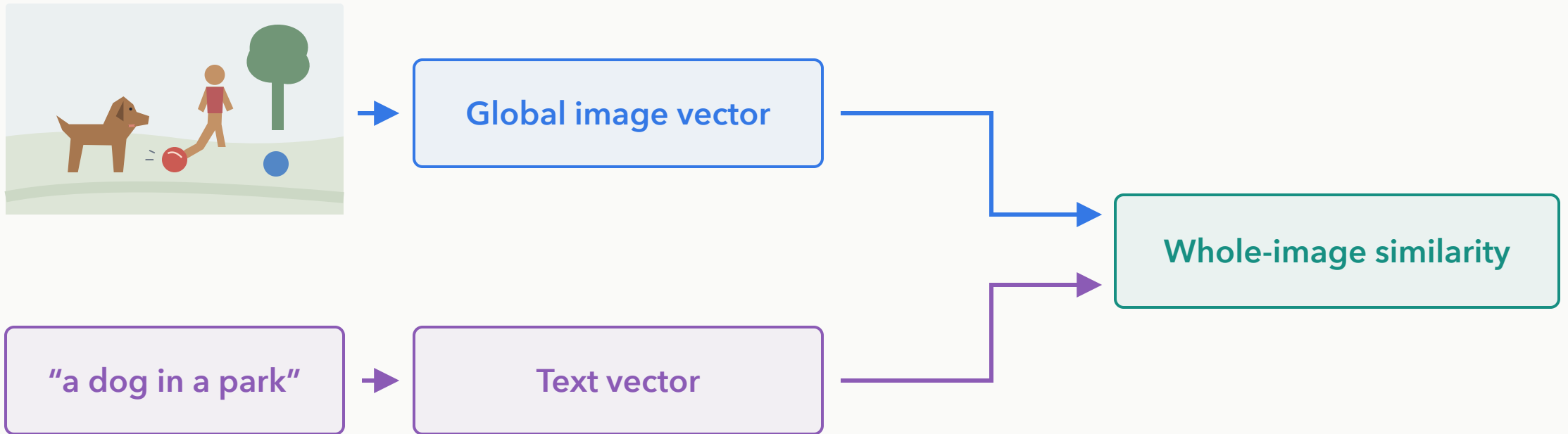
CLIP, BUT SPATIAL

01

CLIP matches a description. How do we answer WHERE?

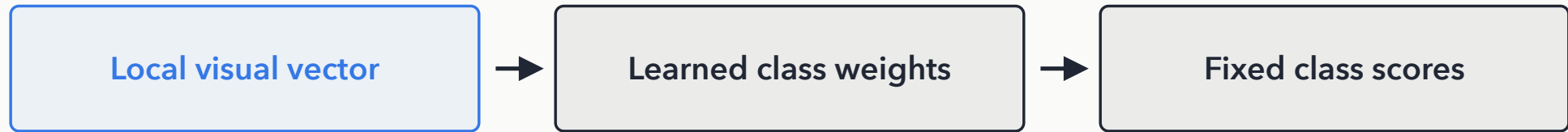


IF is a different question from WHERE



| One global image vector does not specify a box.

Replace the fixed label representation



Supply the class representation through a text encoder.

$$s_{ij} = r_i^\top t_j$$

Text changes which semantic columns we ask the detector to score.

One dot product becomes one matrix cell

$$\begin{aligned} r_1 &= [0.6, 0.8], & t_{\text{dog}} &= [1, 0] \\ s_{1,\text{dog}} &= r_1^\top t_{\text{dog}} \\ &= 0.6 \times 1 + 0.8 \times 0 \\ &= 0.6 \end{aligned}$$

row 1 · dog query

0.6

Normalized, illustrative vectors. These are not measured park features.

▮ The scalar 0.6 answers one region-text comparison.

Build the matrix from comparisons

$$\begin{aligned}
 t_{\text{ball}} &= [0, 1] \\
 s_{1,\text{ball}} &= 0.6(0) + 0.8(1) = 0.8 \\
 r_2 &= [0.8, 0.6] \Rightarrow [0.8, 0.6] \\
 r_3 &= [-1, 0] \Rightarrow [-1, 0]
 \end{aligned}$$

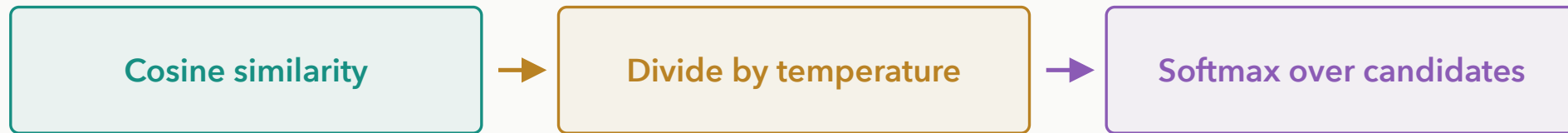
	dog	ball
r_1	0.6	0.8
r_2	0.8	0.6
r_3	-1.0	0.0

$$R \in \mathbb{R}^{N \times d}, \quad T \in \mathbb{R}^{M \times d}, \quad S = RT^\top$$

ROW: which text fits this region? COLUMN: which region fits this text?

Rows index candidate regions. Columns index supplied text queries.

Similarity, logit and probability are distinct



$$s_{ij} = r_i^\top t_j, \quad z_{ij} = s_{ij} / \tau, \quad p(j \mid i) = \frac{e^{z_{ij}}}{\sum_k e^{z_{ik}}}$$

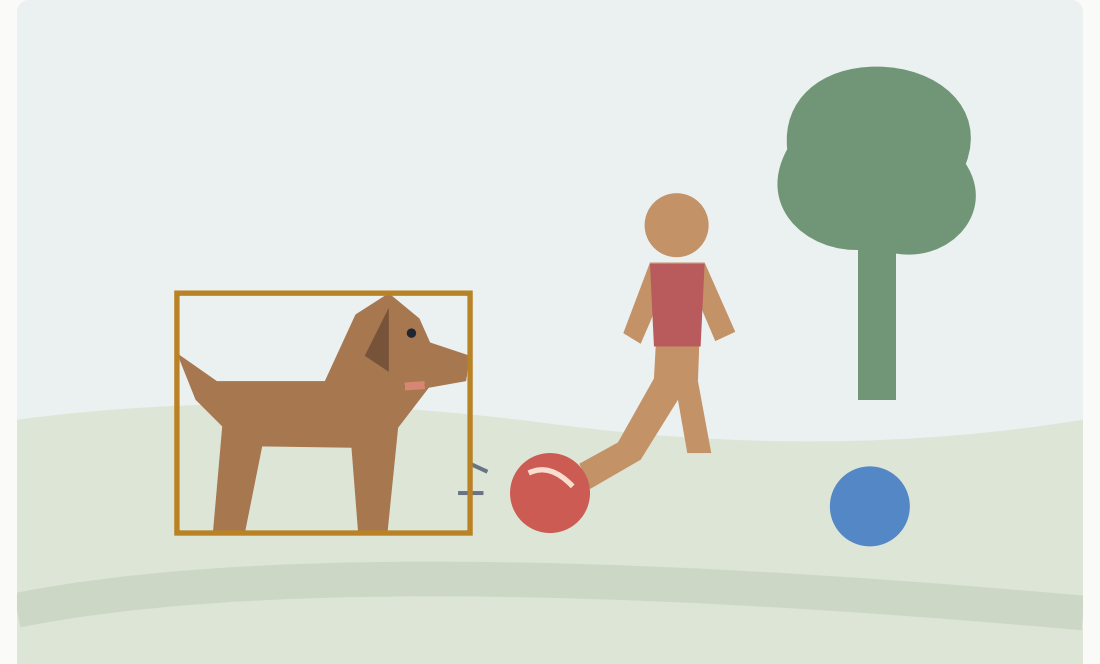
This softmax is a teaching example. Detectors may instead use independent sigmoid scores.

| A cosine score is not automatically a calibrated confidence.

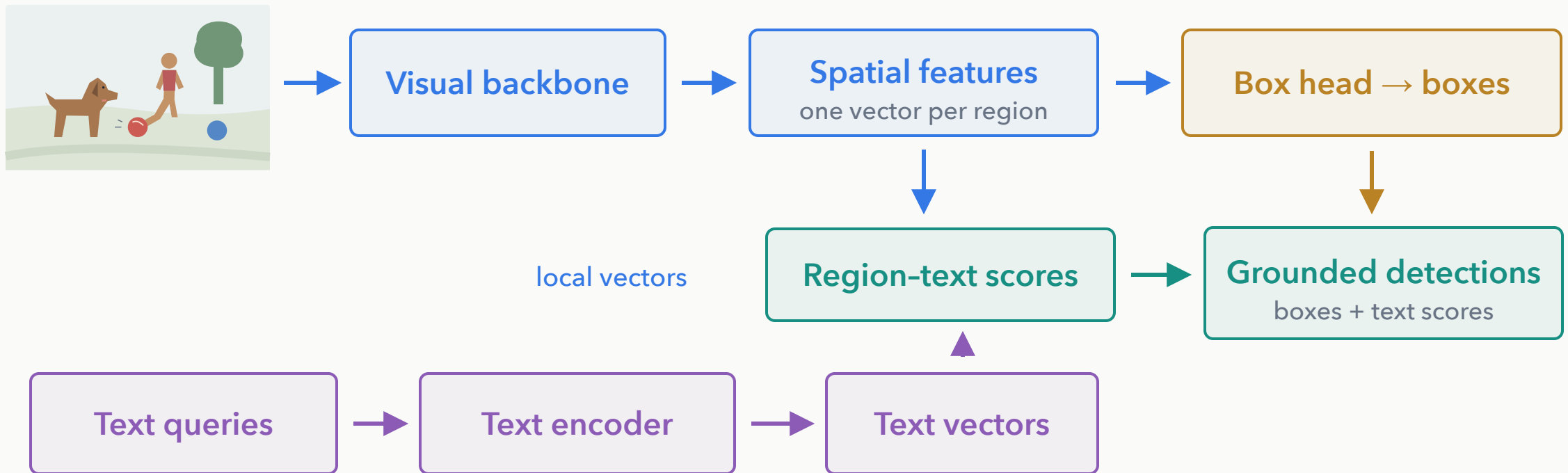
HOW DOES TEXT BECOME A DETECTION?

02

Keep the spatial features. Add geometry. Match text.

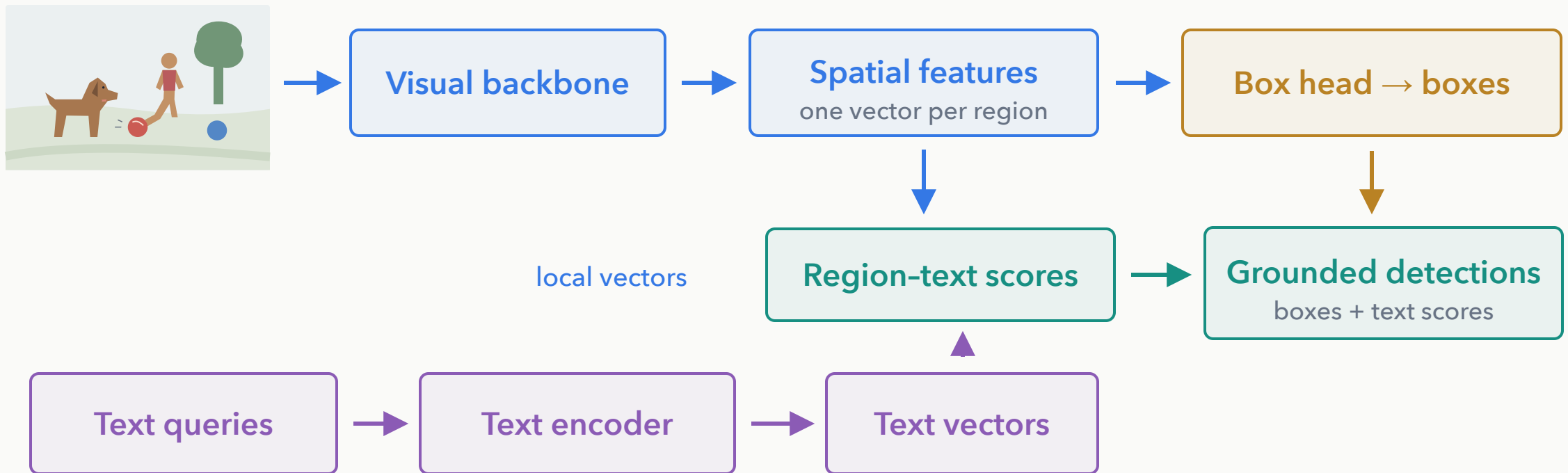


Derive a text-defined detector



CLIP-like semantics + spatial features + box prediction.

Example: OWL-ViT-style detection

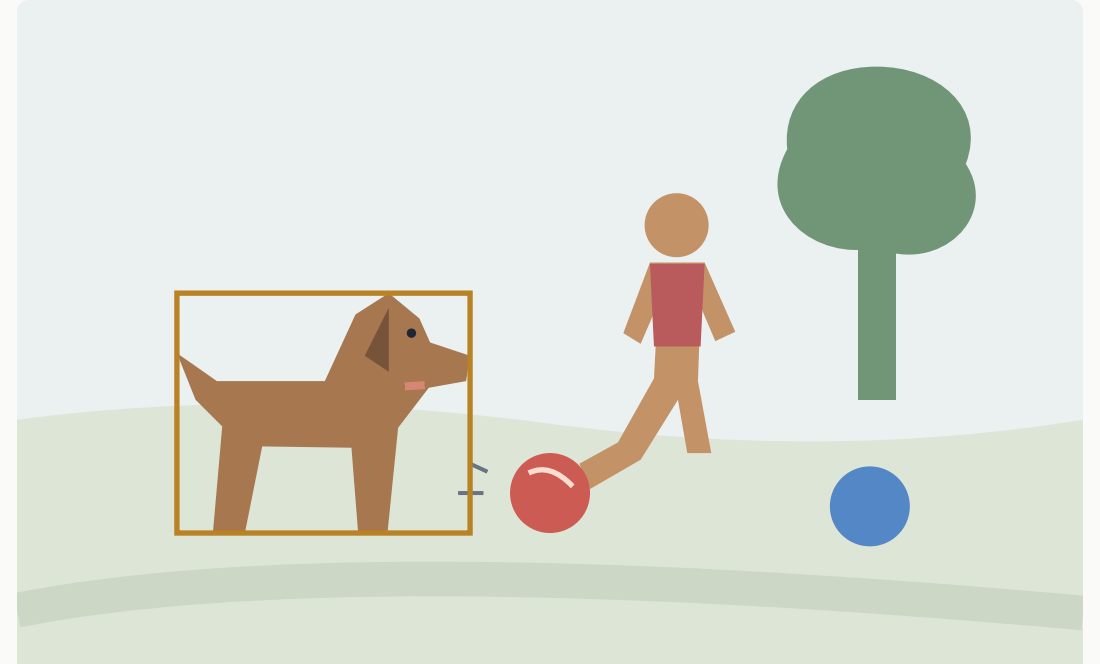


| The local features are trained for detection; this is more than global CLIP retrieval.

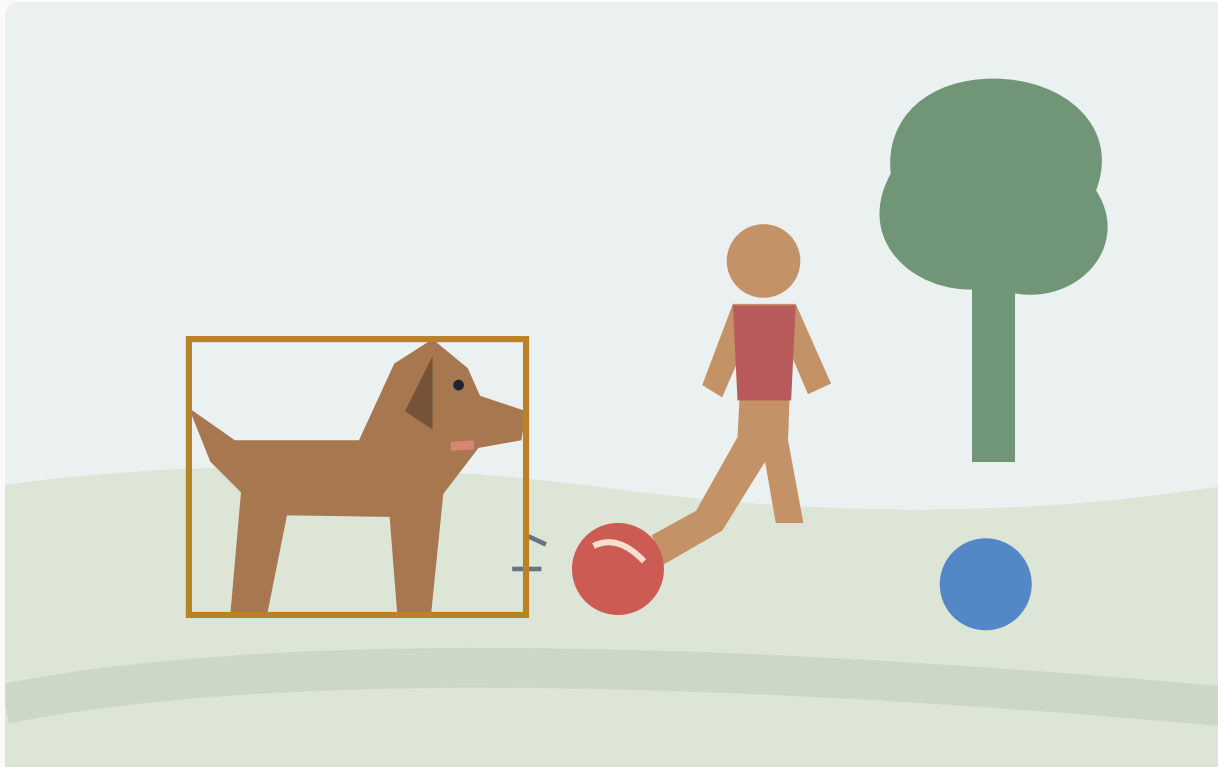
FROM CATEGORY TO REFERRING EXPRESSION

03

Which particular instance does the phrase mean?



A description can add constraints



"dog"

"brown dog"

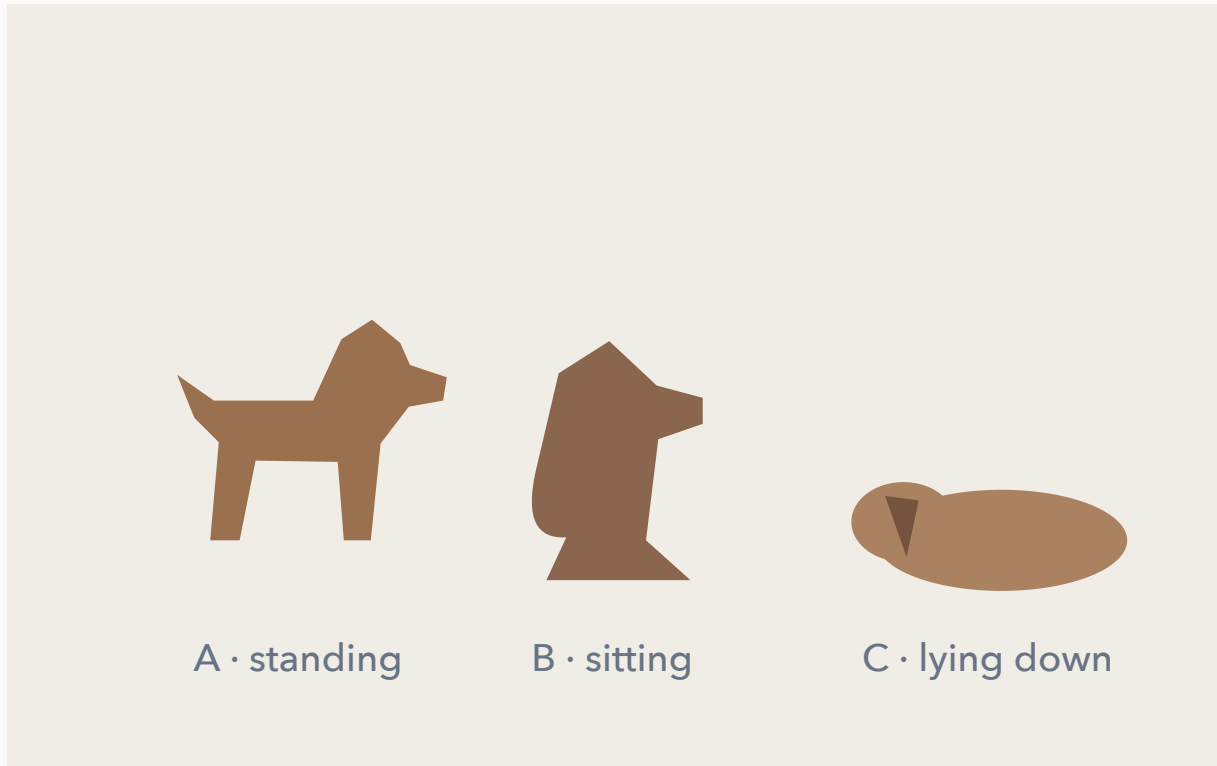
"dog beside the red ball"

"dog beside the child"

CATEGORY → ATTRIBUTE → RELATION

Phrase grounding links language to the region it refers to.

Make the alternatives plausible



“dog”

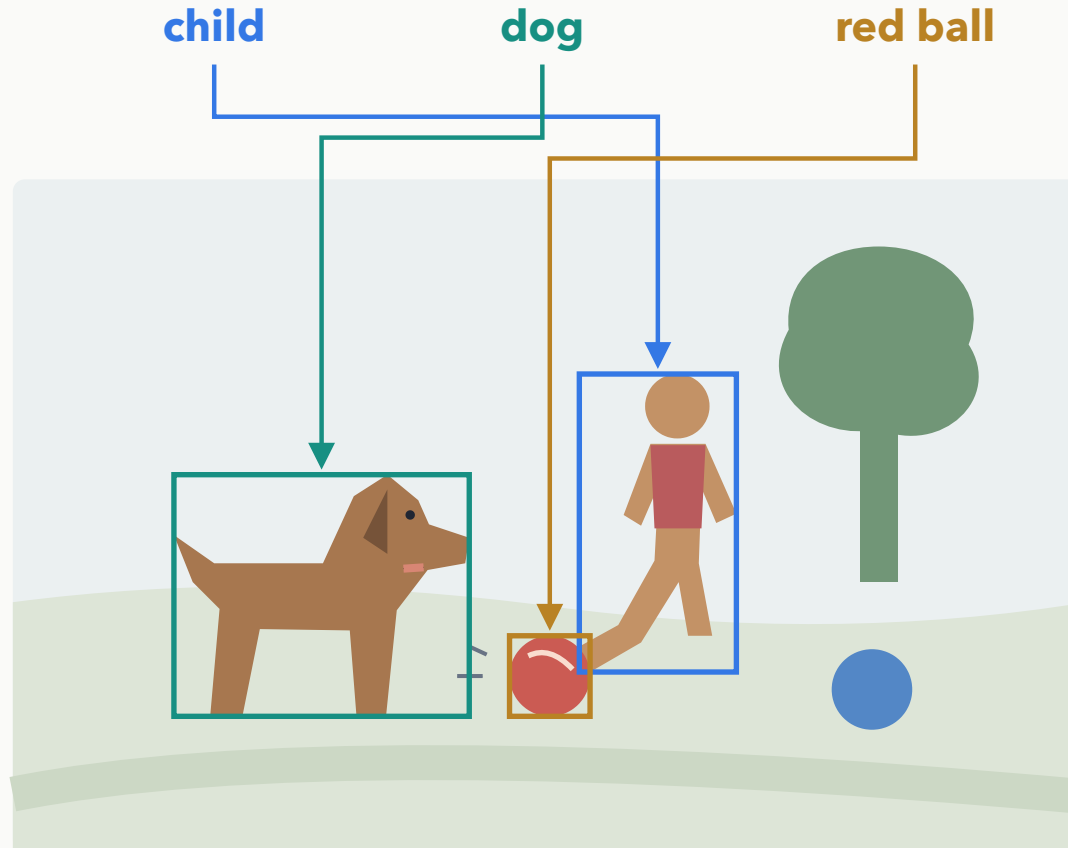
Three valid instances.

“the sitting dog in the middle”

B · posture and position select it.

| A useful grounding test includes competing instances of the same class.

"A child stands near a dog and a red ball."

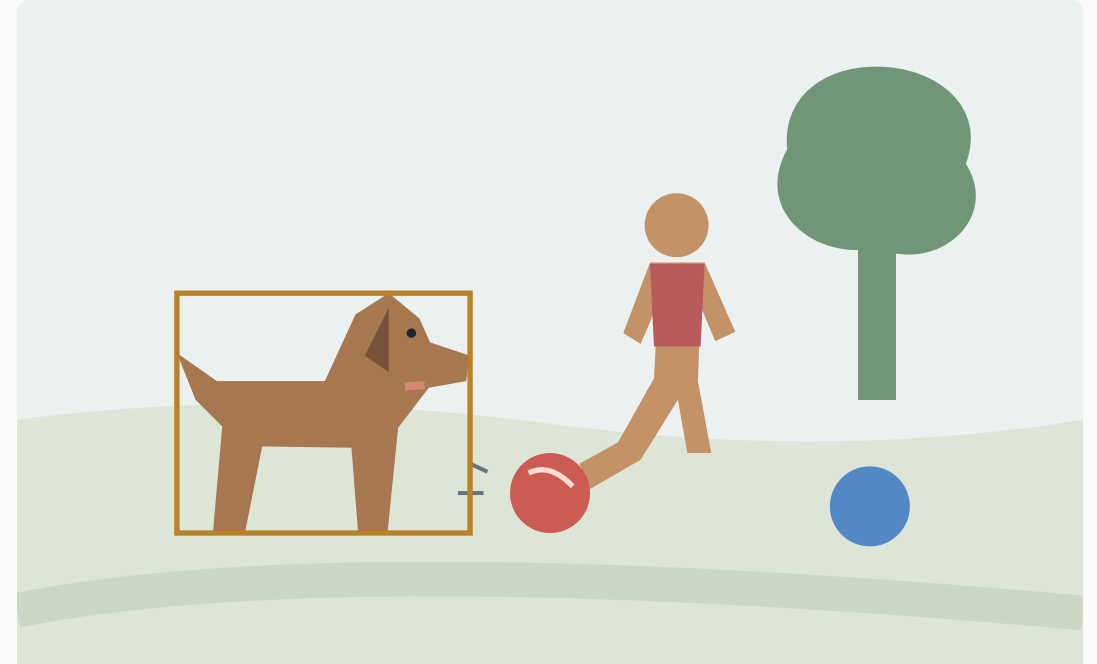


| Language spans and spatial regions need explicit correspondence.

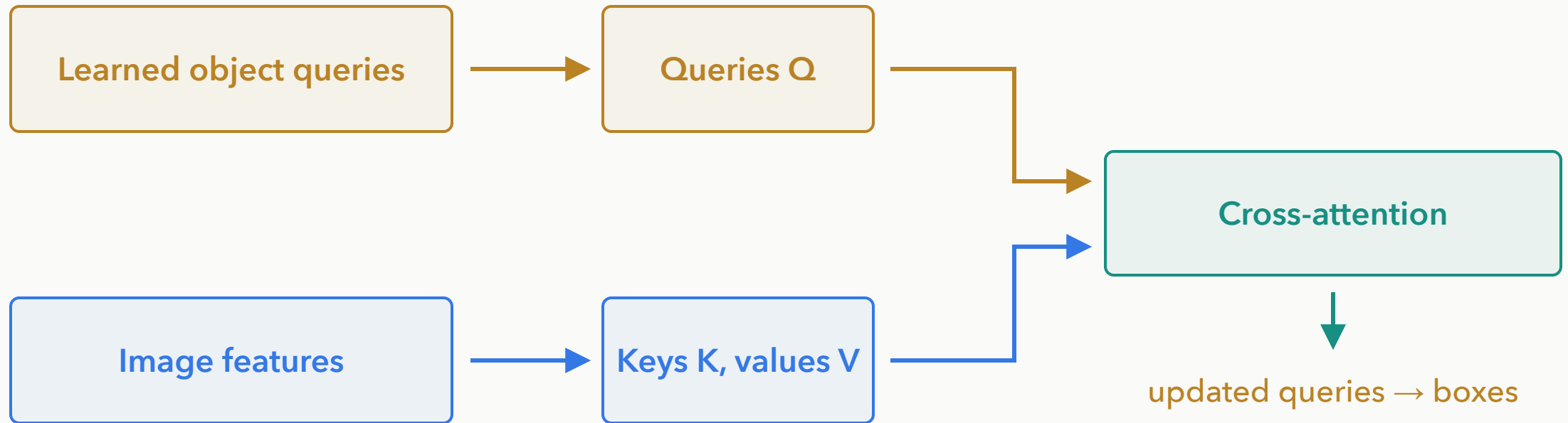
LANGUAGE-GUIDED OBJECT QUERIES

04

What if language could affect which queries are useful?

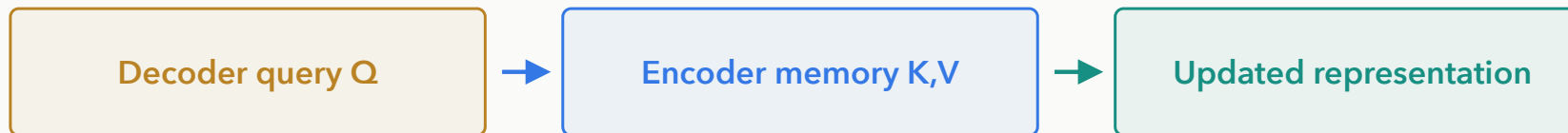


Recall DETR: queries read the image



| A query is a learned request for information, not a preassigned object name.

The same operation, a spatial question



Earlier: generate the next word.

Here: locate what a phrase refers to.

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$$

| Beyond Attention → object queries reading visual and multimodal memories.

One query mixes three values

$$q = [1, 0], \quad K = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$qK^\top = [2, 0, -2]$$

$$z = qK^\top / \sqrt{2} = [1.414, 0, -1.414]$$

Key locations: dog · red ball · background

$$\alpha = \text{softmax}(z) \approx [0.768, 0.187, 0.045]$$

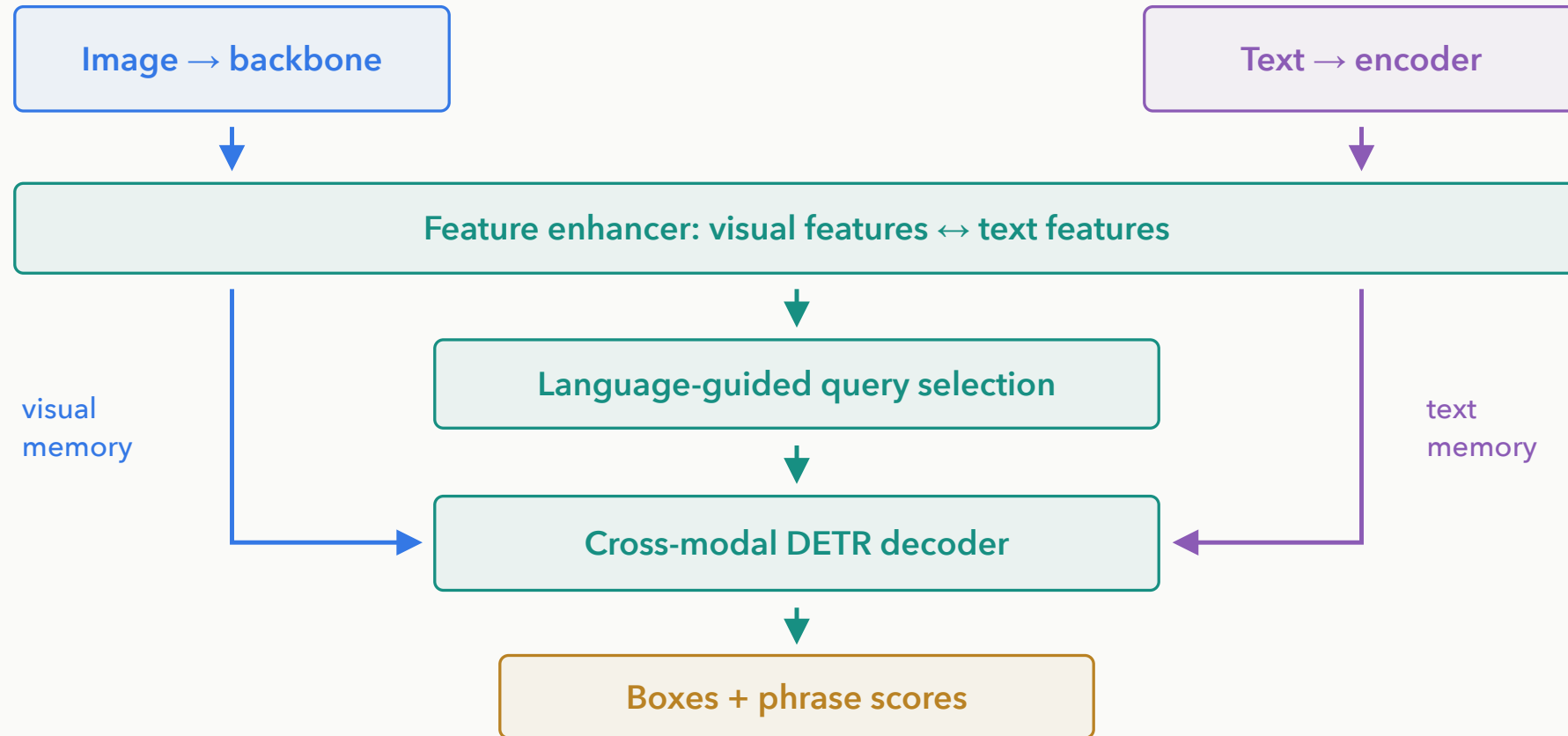
$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\alpha V \approx [0.768, 0.187]$$

ILLUSTRATIVE ATTENTION WORKSHEET · spatial labels locate toy keys, not hidden semantics.

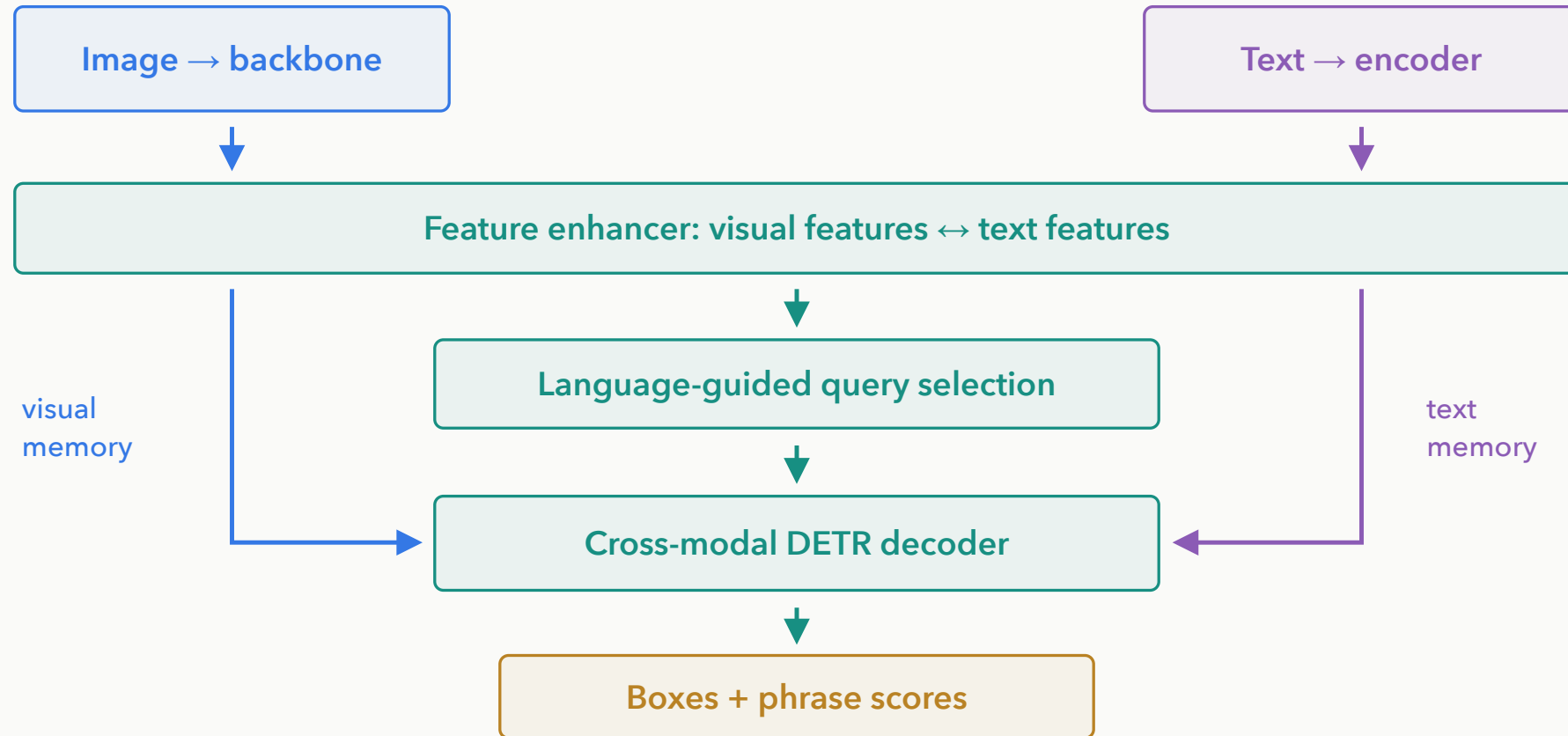
Attention produces a value mixture; a downstream head still has to predict geometry.

Let text influence feature selection



Language can influence features, selected queries and decoder updates.

This is the Grounding DINO design

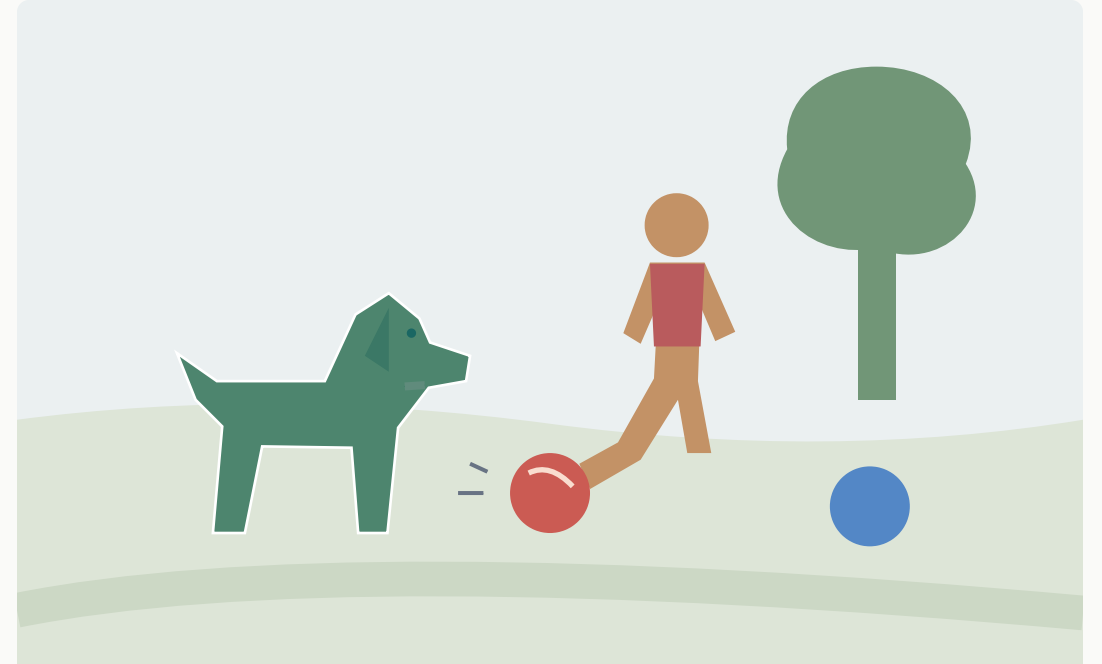


A language-conditioned detector, with boxes and text alignment as outputs.

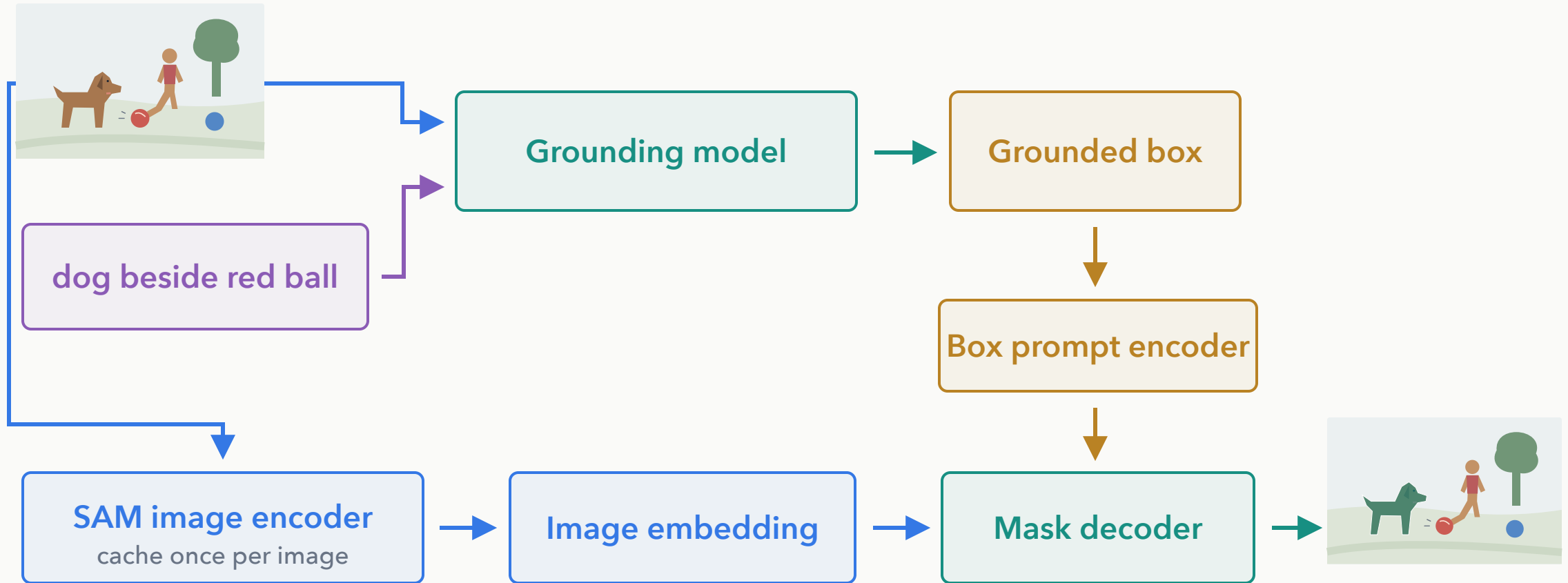
TEXT → BOX → MASK

05

Language selects a region. How do we recover its pixels?

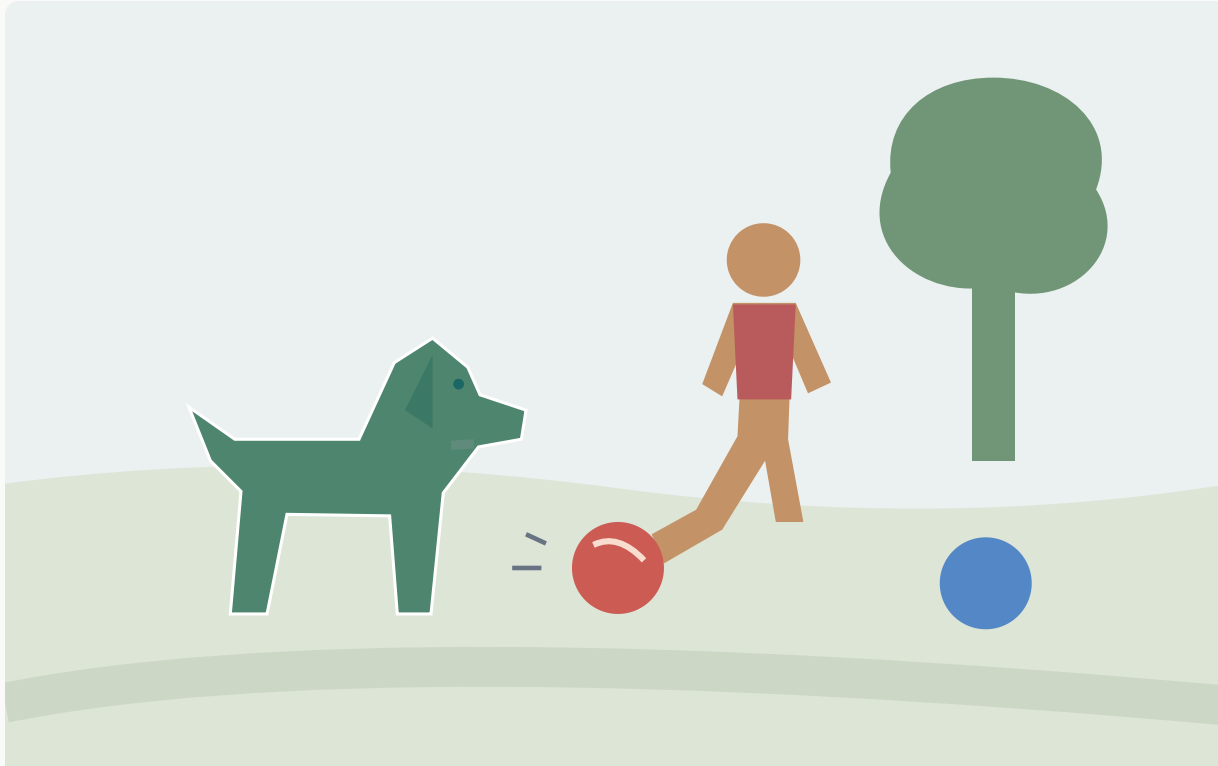


Compose a grounder with a segmenter



Language selects the region. SAM recovers its precise shape.

The user asks for a mask



“segment the dog beside the red ball”

One requested region.

One pixel mask.

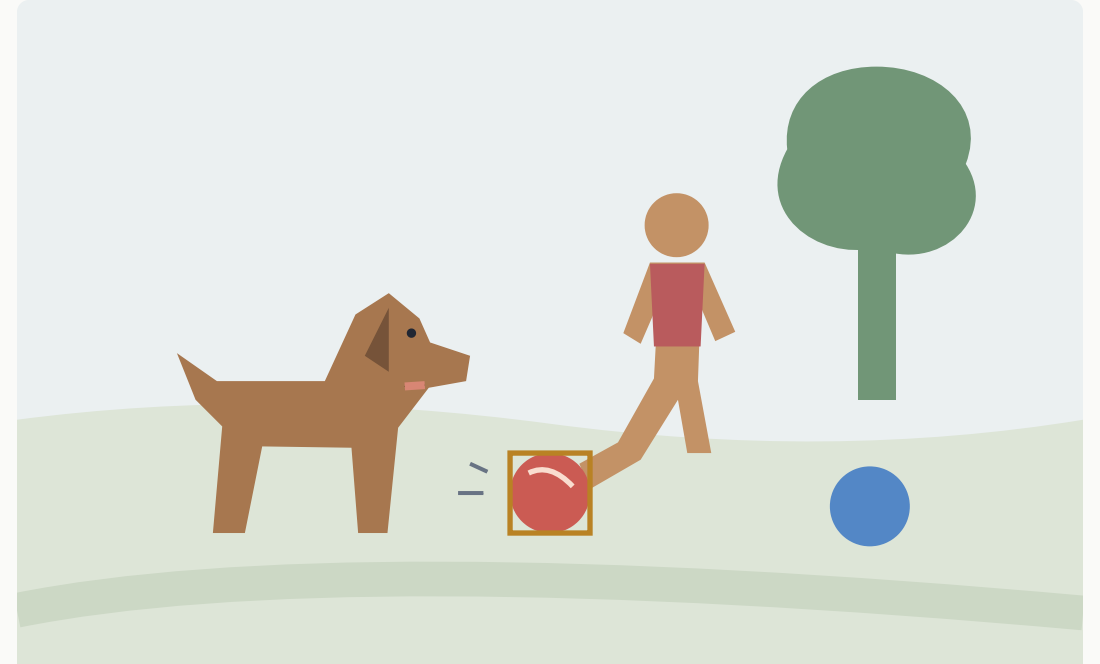
Intermediate boxes can disappear from the user’s view.

| A clean mask can still belong to the wrong instance.

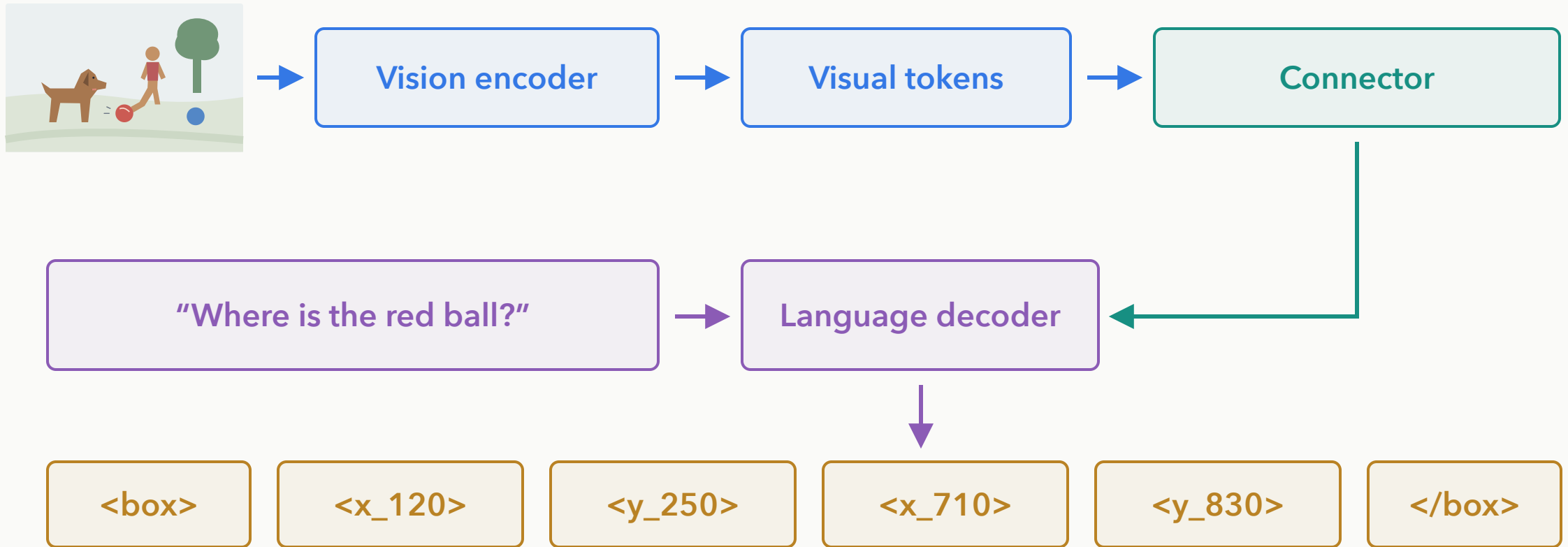
CAN A VLM GENERATE SPACE?

06

What if the answer is a spatial sequence?



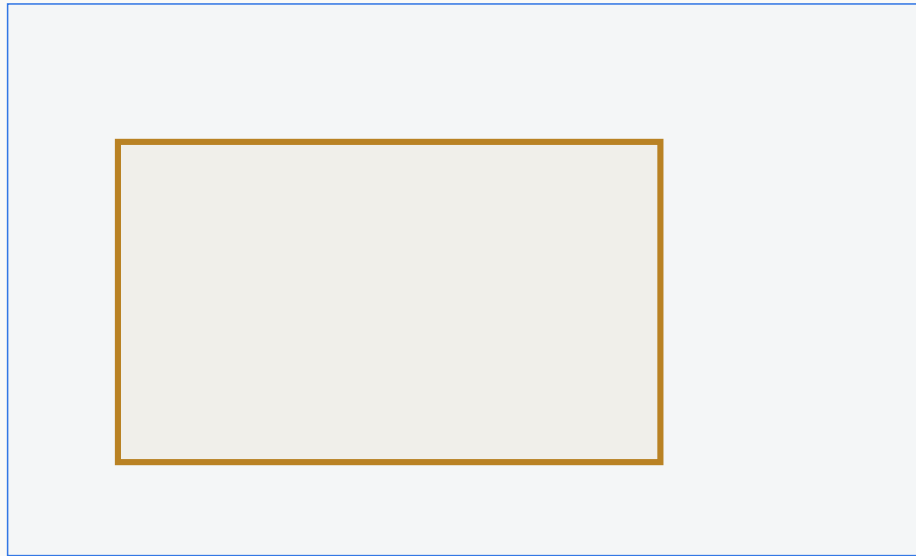
A familiar VLM, a different answer grammar



Illustrative coordinate grammar and toy box · not a measured answer

Dedicated head: box coordinates. Sequence decoder: spatial tokens.

Continuous coordinate $\rightarrow$ bin $\rightarrow$ token



normalized box (0.12, 0.25, 0.71, 0.83)

$$x_1 = .12 \quad \longrightarrow \quad \lfloor 1000(.12) \rfloor = 120 \quad \longrightarrow \quad \langle x_{120} \rangle$$

$$y_1 = .25 \quad \longrightarrow \quad \lfloor 1000(.25) \rfloor = 250 \quad \longrightarrow \quad \langle y_{250} \rangle$$

$$x_2 = .71 \quad \longrightarrow \quad \lfloor 1000(.71) \rfloor = 710 \quad \longrightarrow \quad \langle x_{710} \rangle$$

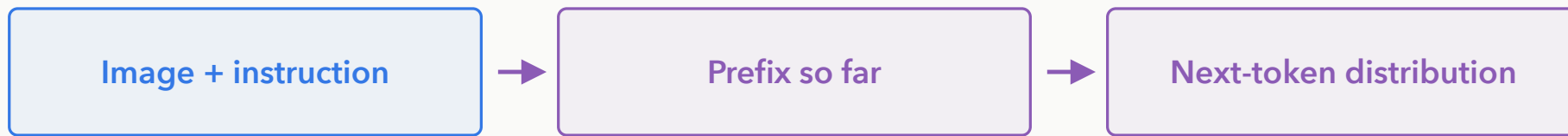
$$y_2 = .83 \quad \longrightarrow \quad \lfloor 1000(.83) \rfloor = 830 \quad \longrightarrow \quad \langle y_{830} \rangle$$

Toy convention: 1,000 bins/axis, floor then clip to 999.

Decode at bin centres.

Each coordinate becomes a discrete symbol under a declared convention.

Predict one spatial token at a time



$$p(y \mid I, q) = \prod_t p(y_t \mid y_{<t}, I, q)$$

A valid sequence must also describe the correct region.

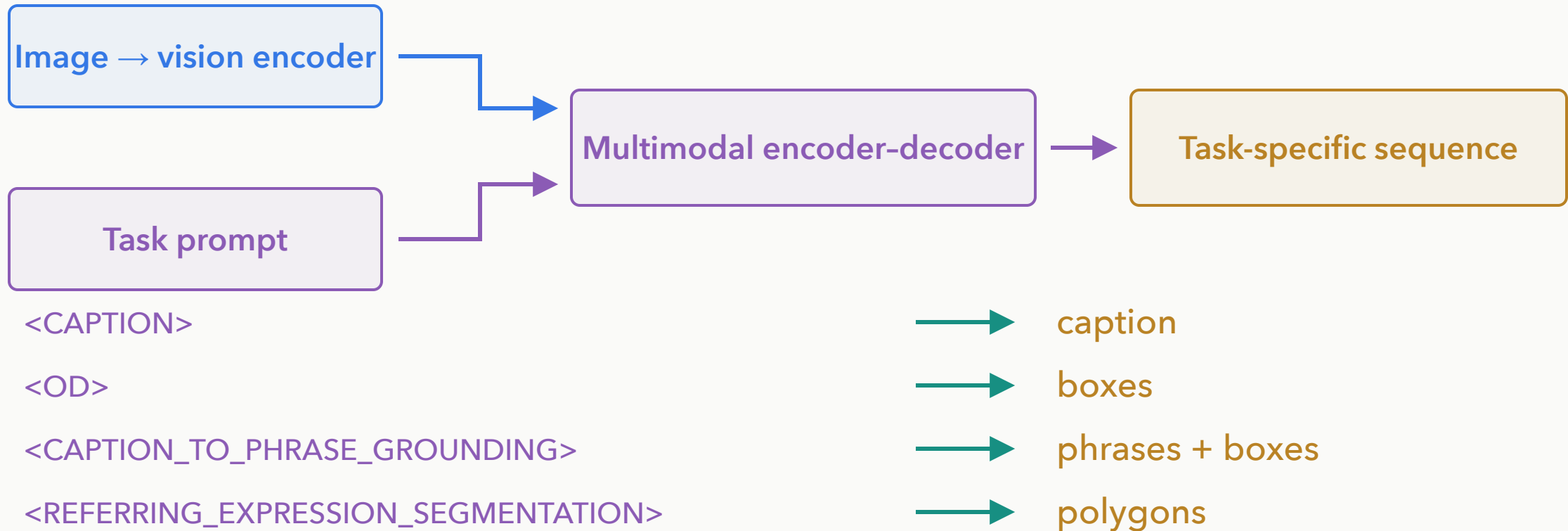
Example: Kosmos-2 grounds generated phrases



1,024 location tokens · each token represents a grid cell

| Spatial grammars are model-specific: Kosmos-2 uses two corner tokens.

Example: Florence-2 changes tasks through prompts



The prompt selects an output grammar; a parser turns spatial tokens into geometry.

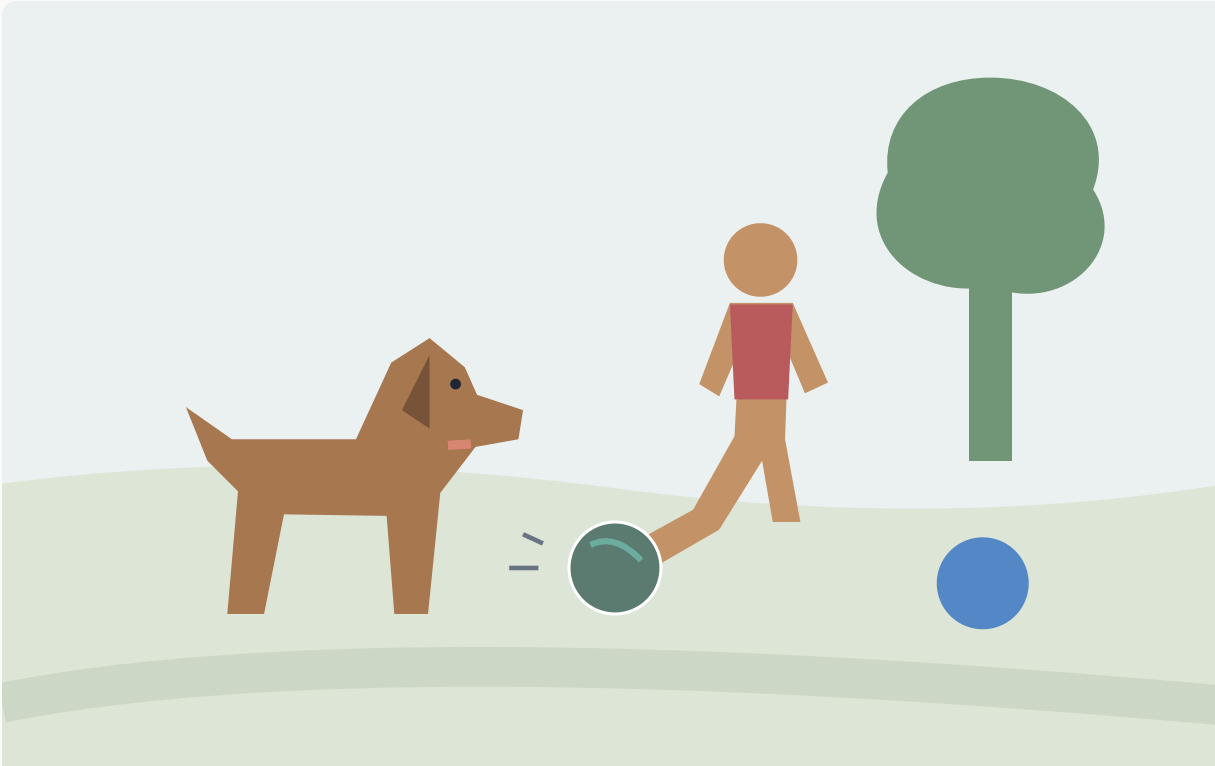
CAN A VLM REASON INTO A MASK?

07

The target may be implicit in the instruction.



From naming to interpreting the target



"segment the dog"

"segment the dog beside the red ball"

"segment the object the child is playing with"

Reference target: red ball.

Visible cue: the child's foot meets the ball.

| The instruction can describe a role without naming the object.

The request can depend on world knowledge



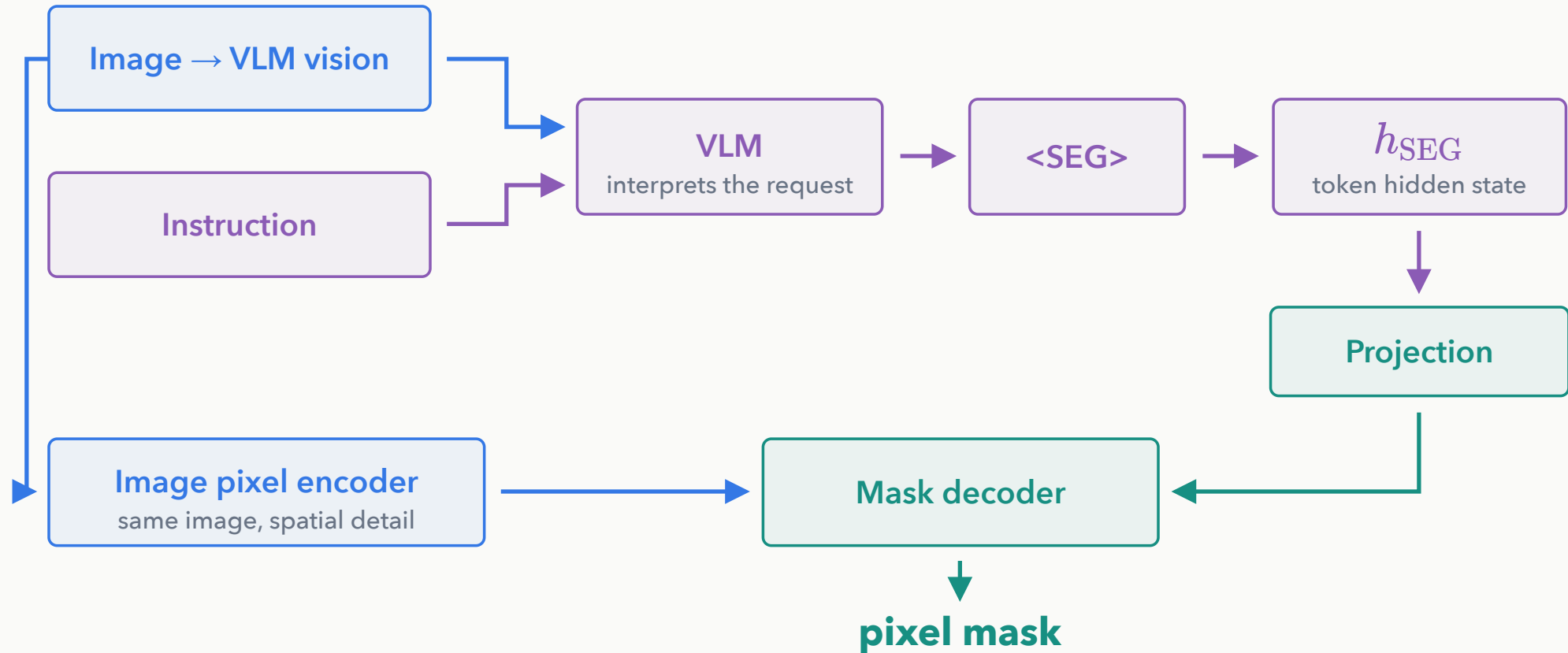
"segment the object used to stay dry in rain"

Umbrella

A second controlled illustration.
No umbrella is hidden in the park scene.

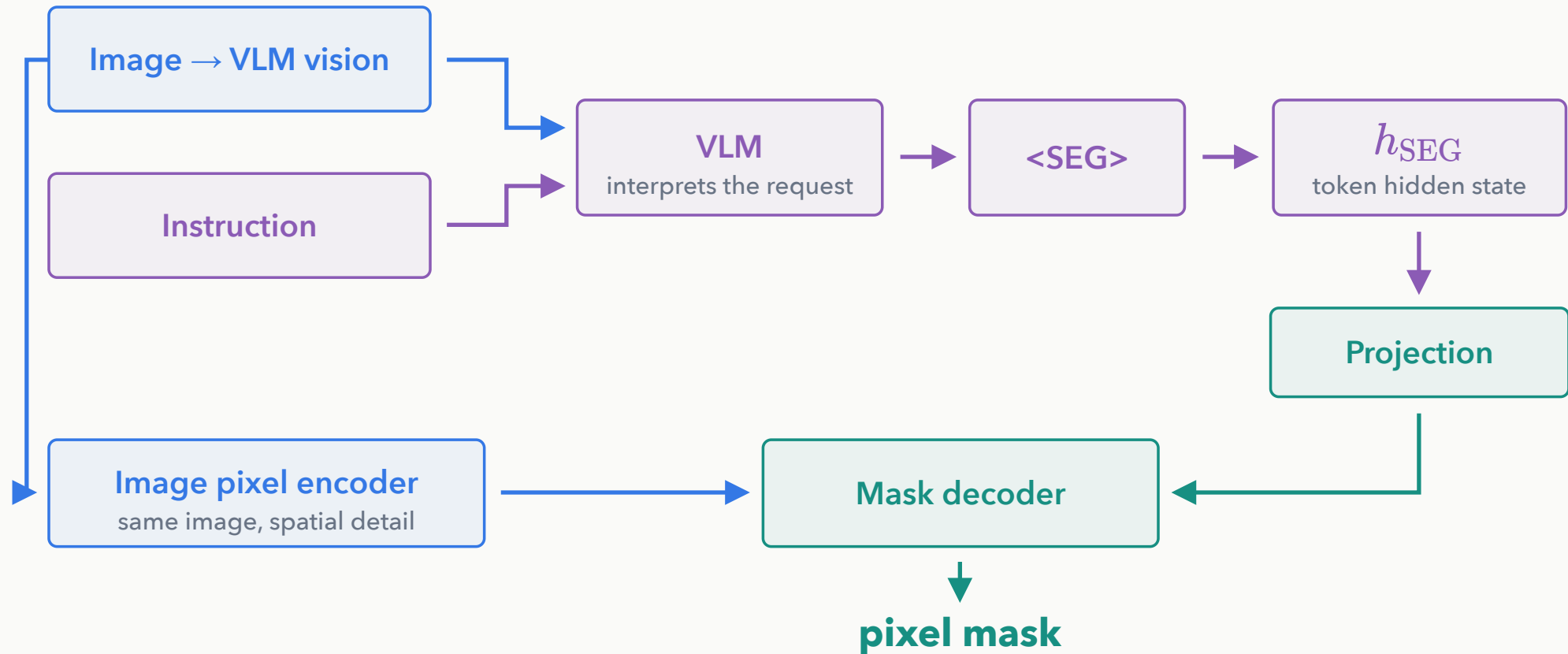
| Interpret the request, then identify the pixels of the chosen object.

Turn a language representation into a mask



Language interprets WHAT is meant; the mask decoder predicts WHICH PIXELS.

Example: LISA-style reasoning segmentation

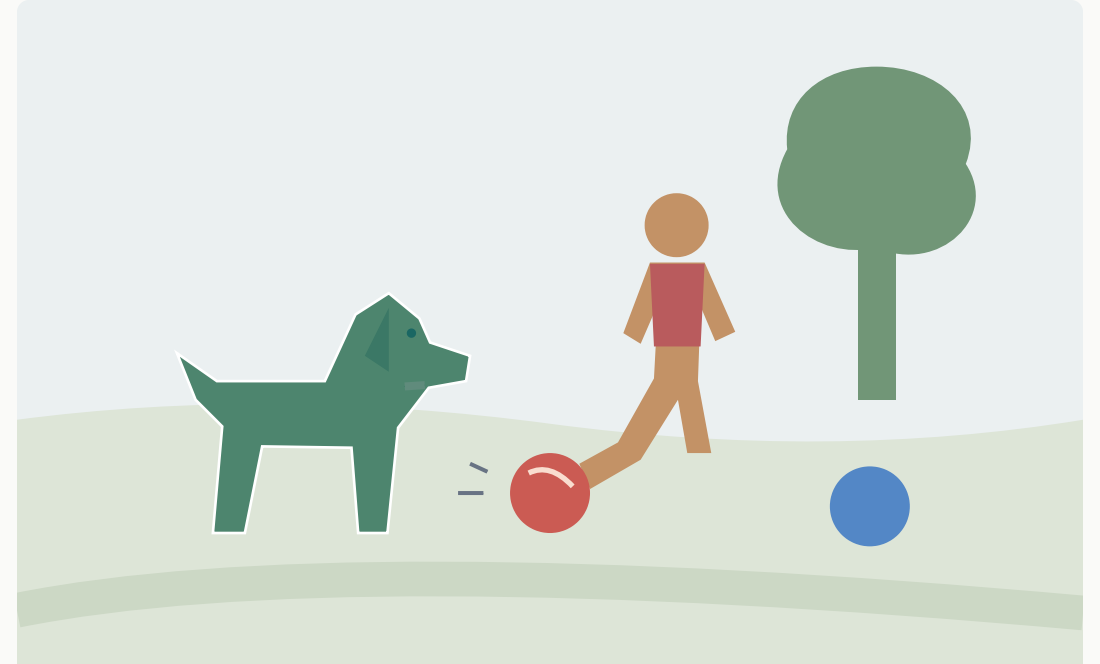


A learned segmentation representation connects a VLM to a pixel decoder.

FOUR ARCHITECTURES FOR LANGUAGE-GROUNDED VISION

08

Compare where language enters, what leaves, and what is trained.



Four ways to make language spatial

Open vocabulary

text vectors → matching



boxes + scores
alignment + box

Phrase grounding

text ↔ features / queries



referred boxes
alignment + box

Ground + segment

text → box → SAM



pixel masks
grinder + mask

Generative / reasoning

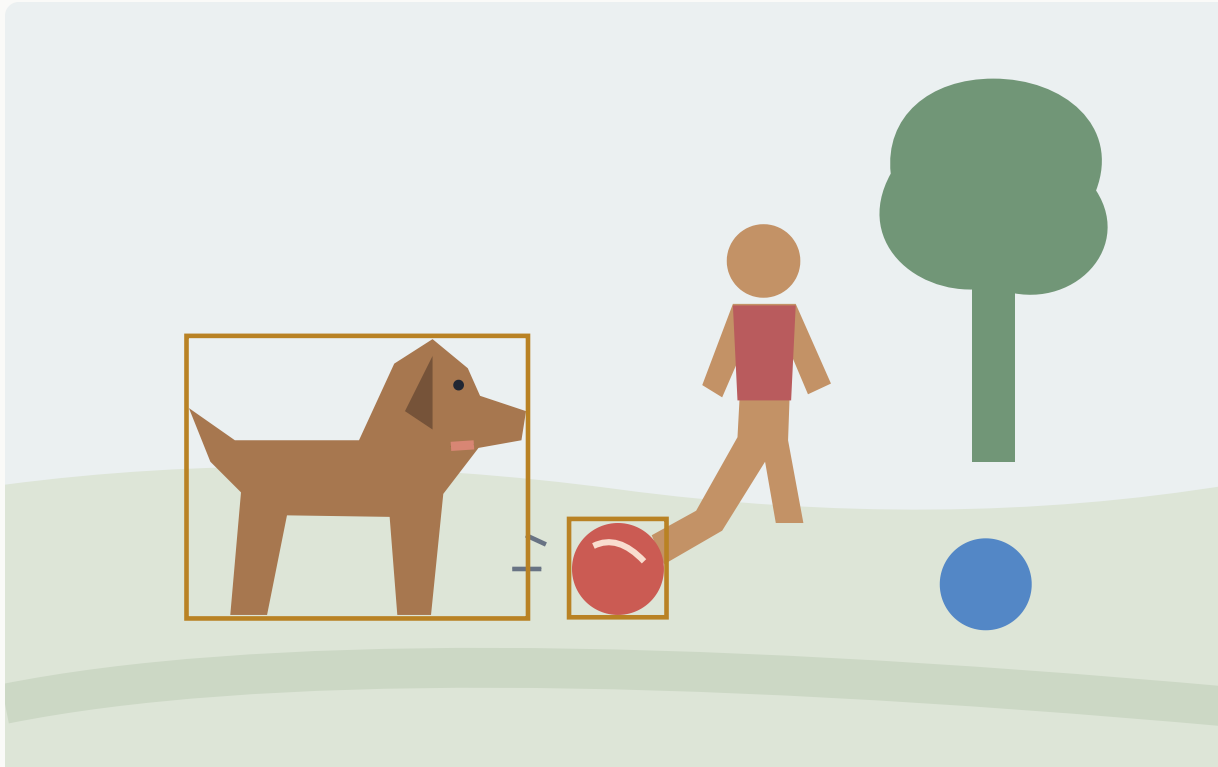
instruction → VLM



tokens OR SEG → mask
token OR LM + mask

Language entry → output representation → learning signal.

A recorded answer can miss the intended target



"ball beside the dog."

dog · 0.603

ball · 0.476

Recorded alignment scores, not accuracy.

The requested target is the ball.

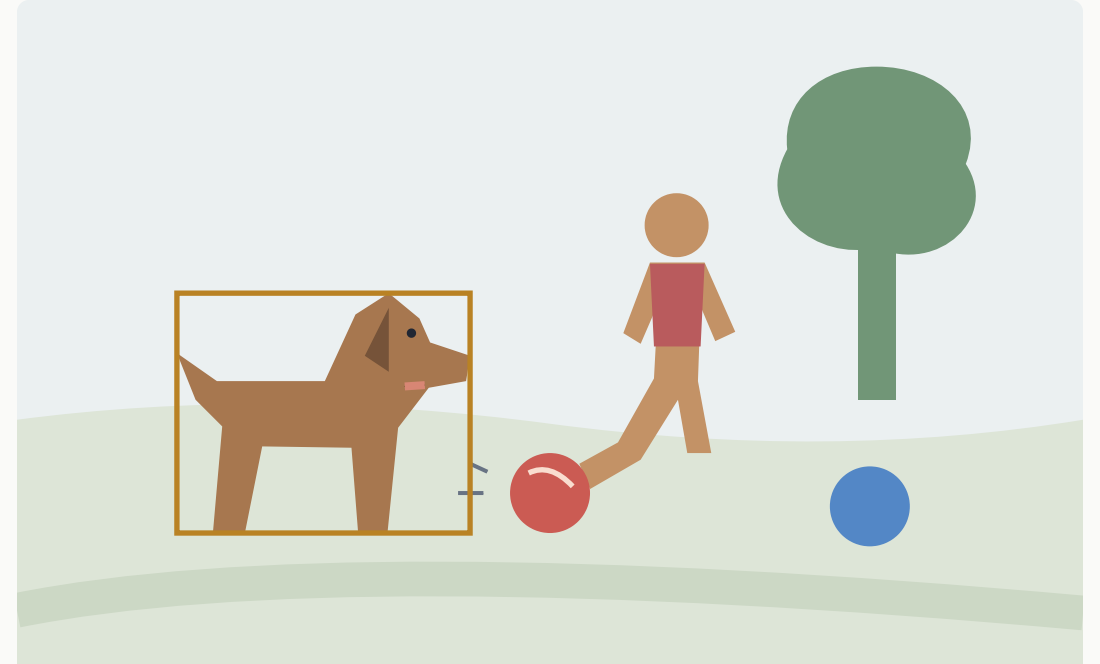
[Explore all seven recorded runs ↗](#)

Keep the raw predictions. Inspect which words and regions they link.

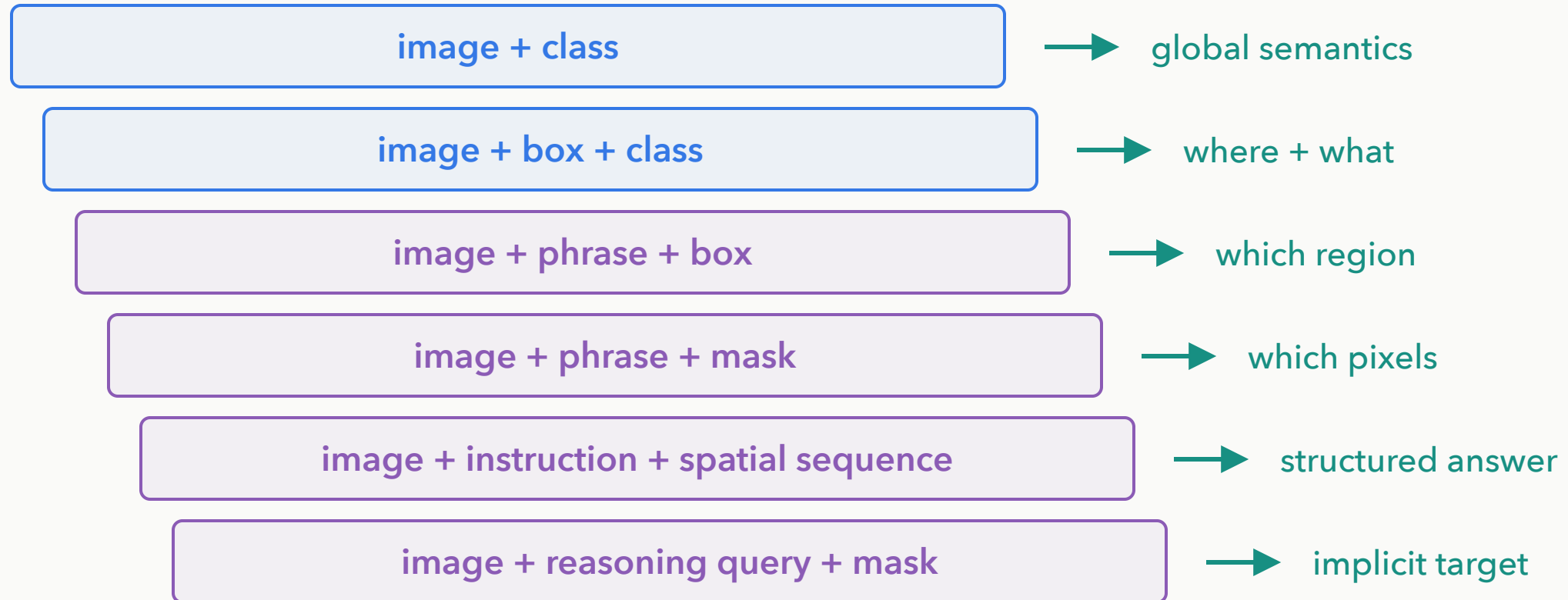
TRAINING / EVALUATION SYNTHESIS

09

The requested output determines supervision and evaluation.



What annotation teaches each behavior?



| Richer instructions require targets that make their intended region identifiable.

Loss follows the output representation

Box detector

$$\mathcal{L}_{\text{align}} + \lambda_b \mathcal{L}_{\text{box}}$$

Mask decoder

$$\mathcal{L}_{\text{mask}}$$

Generated coordinates

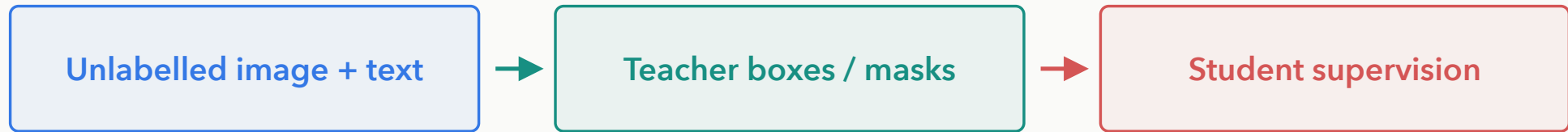
$$\mathcal{L}_{\text{LM}} = - \sum_t \log p(y_t \mid y_{<t}, I, q)$$

VLM + mask

$$\mathcal{L}_{\text{LM}} + \lambda_m \mathcal{L}_{\text{mask}}$$

Train the semantic choice and the spatial answer the system actually produces.

A teacher can scale labels—and its mistakes

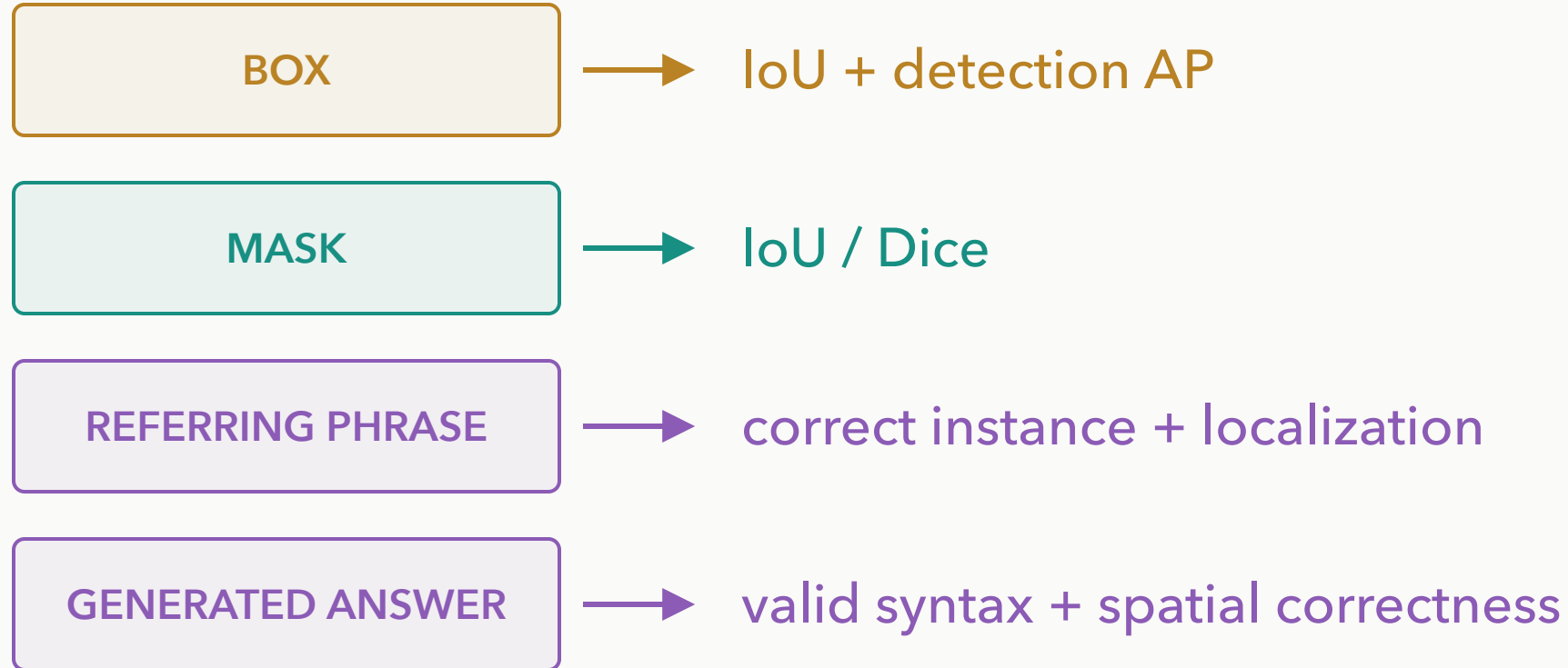


A wrong instance can become a very precise training target.

Audit relations, absence and domain shift. Keep a human-checked held-out set.

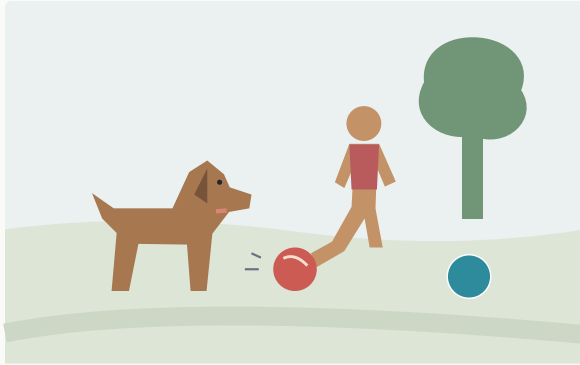
| More pseudo-labels do not automatically mean better grounding.

Evaluate the output and the target identity

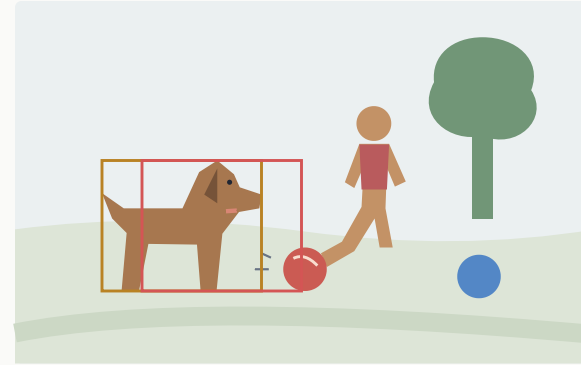


State the matching rule, threshold and split; one metric cannot test every failure.

Separate four failure modes



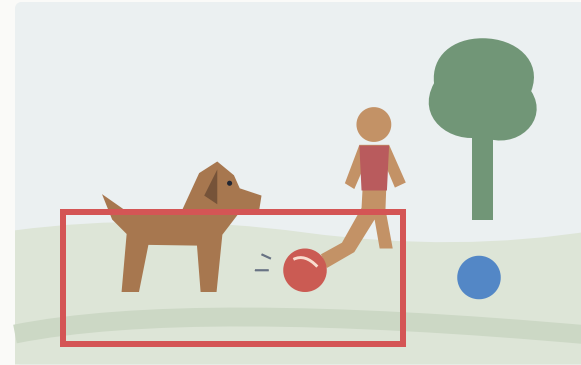
right shape
wrong instance



right object
shifted coordinates



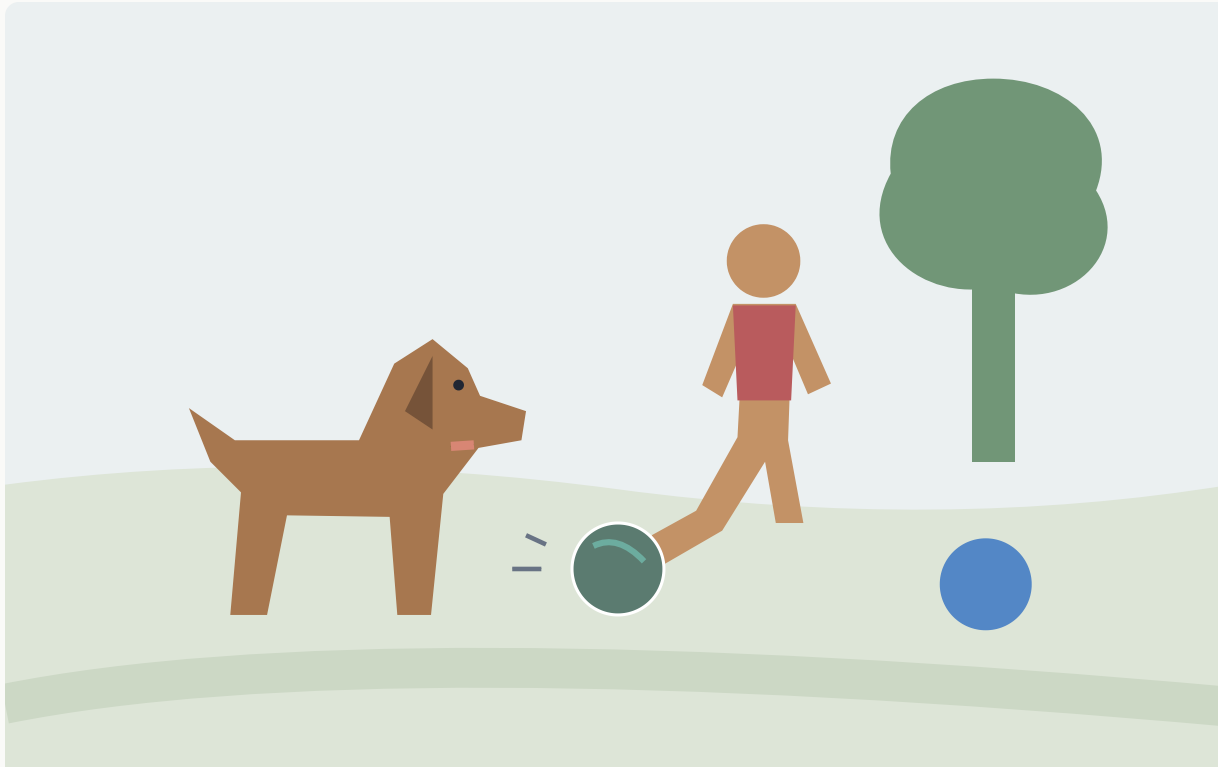
right box
bad mask



absent bicycle
hallucinated

Diagnose target choice, coordinate mapping, shape and absence separately.

Choose a route; explain the representation



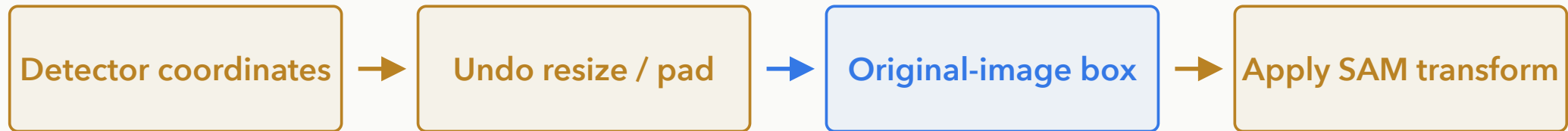
“segment the object the child is playing with”

Where does language enter?
What spatial representation leaves?
What target and loss teach it?

Selection → representation → pixels

Language changes the target. Architecture determines how that choice becomes space.

Coordinate handoff: the same box in two systems



$$(x, y)_{\text{orig}} = ((x_{\text{det}} - p_x) / s_d, (y_{\text{det}} - p_y) / s_d)$$

SAM mask postprocessing: undo SAM resize/padding → original-image mask.

Box format and coordinate frame must be explicit at the interface.

Overlap arithmetic belongs in backup



Two 200×200 boxes

$$\text{Intersection} = 150 \times 200 = 30,000$$

$$\text{Union} = 40,000 + 40,000 - 30,000$$

$$\text{IoU} = 30,000 / 50,000 = 0.60$$

$$\text{Dice} = 60,000 / 80,000 = 0.75$$

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|} = \frac{30000}{50000} = 0.60, \quad \text{Dice} = \frac{2|A \cap B|}{|A| + |B|} = 0.75$$

For nonempty binary masks, $\text{Dice} = 2 \text{IoU} / (1 + \text{IoU})$.

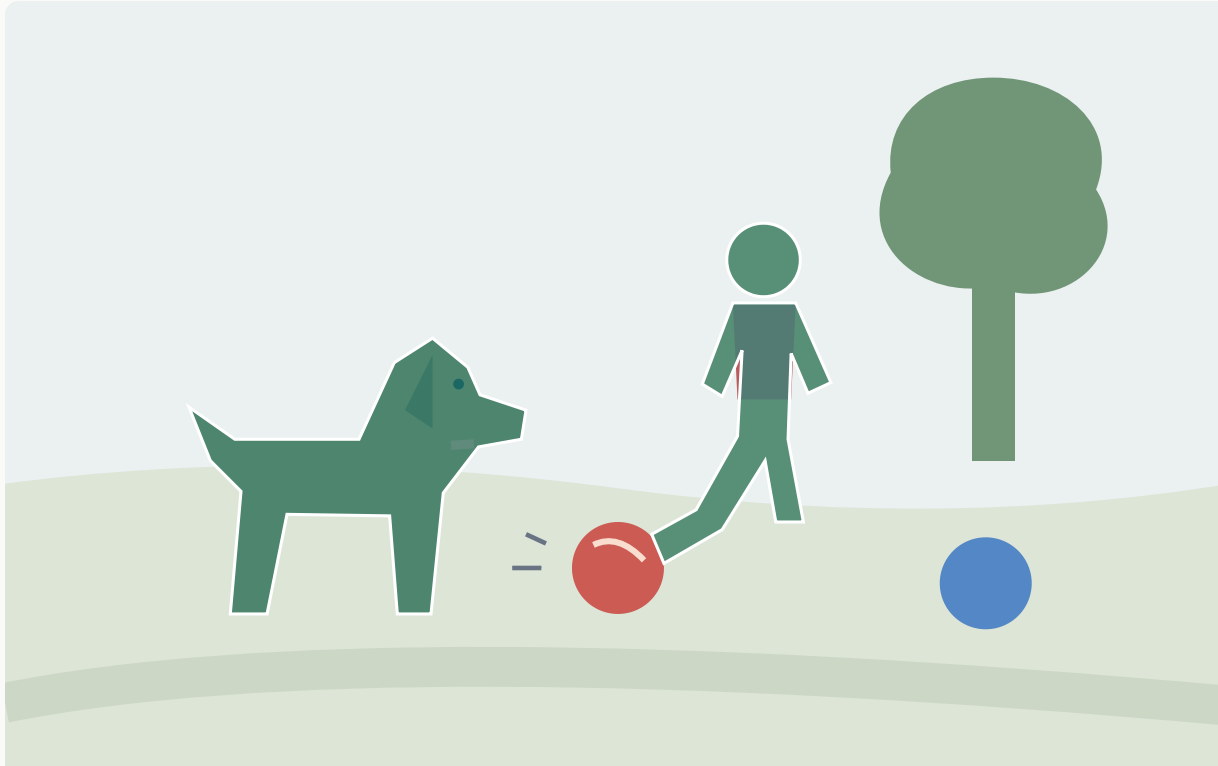
Pseudo-labels need an error audit



Keep teacher version, threshold, prompt, image transform and human-audit split.

| Confidence filtering cannot repair systematic target confusion.

Optional: language can point into masks



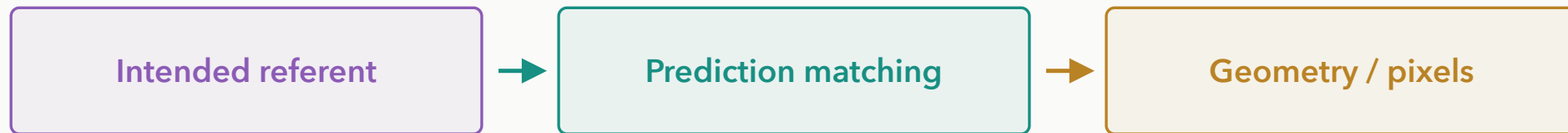
"The [dog] is beside the [child]."

[dog] → dog mask

[child] → child mask

| GLaMM connects grounded language and region masks within conversation.

Specify the evaluation protocol



Box AP: thresholds + class matching + ranked detections.

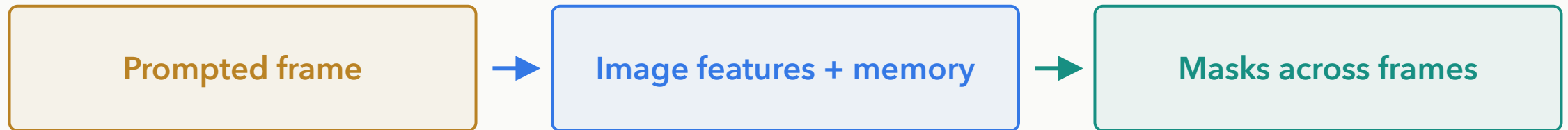
Referring accuracy: target identity + chosen IoU threshold.

Mask overlap: per-object / per-image / global aggregation.

Absence: false positives on negative queries.

| A score without its protocol is not a comparable result.

Optional: carry a spatial prompt through video



SAM 2 extends promptable segmentation to video.
Language selection still depends on the surrounding system.

| Temporal propagation adds a memory problem beyond this image-only lecture.

Primary sources and implementation references

CLIP · 2021

OWL-ViT · 2022

GLIP · 2022

DETR · 2020

Grounding DINO · 2024

Segment Anything · 2023

Grounded SAM · implementation

Kosmos-2 · 2023

| Original diagrams; illustrative examples are labelled separately from recorded inference.

Primary sources and implementation references

Florence-2 · 2024

Florence-2 · task parser

LISA · 2024

LISA · segmentation interface

GLaMM · 2024

COCO · evaluator

SAM 2 · 2024

| Original diagrams; illustrative examples are labelled separately from recorded inference.